Introduction to Quantum Field Theory, Lecture notes

Università degli Studi di Torino
Academic Year 2026/2027
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Lesson 1, 23/09

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Lesson 2, 24/09

1 Natural units

From this point onwards, natural units will be used:

$$\hbar = c = 1$$

Note: this messes up dimensionality, but, conveniently, forces mass and energy to have the same units

$$E = mc^2 \underset{c=1}{\rightarrow} E = m,$$

and length and time to have the same dimensionality

$$x = ct \underset{c=1}{\rightarrow} [x] = [t].$$

To convert back into more familiar units the constants are, respectively,

$$c = 299.792.458 \text{ m/s}$$

$$\hbar c \approx 0.97 \text{ GeV·fm}$$

2 Klein-Gordon Equation

The relativistic quantum mechanics' fundamental equation is

$$(\Box + m^2)\psi(x) = 0$$

where

$$\Box = \partial^\mu \partial_\mu = \frac{\partial^2}{\partial t^2} - \nabla^2$$

and $\psi(x)$ is still the Schrödinger's wavefunction.

It will now be shown that this equation **does not** conserve probability, using a continuity equation.

Regular quantum mechanics states

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0,$$

where

$$\rho = \langle \psi_t | \psi_t \rangle \qquad \mathbf{j} = \frac{i}{m} (\psi^* \boldsymbol{\nabla} \psi - \psi \boldsymbol{\nabla} \psi^*)$$

Consider the Klein-Gordon equation left-multiplied by $\psi^*(x)$ and its conjugate

$$\begin{cases} \psi^*(x) \left(\Box + m^2\right) \psi(x) = 0 \\ \psi(x) \left(\Box + m^2\right) \psi^*(x) = 0 \end{cases}$$

their difference is

$$\psi^*(\Box + m^2)\psi - \psi(\Box + m^2)\psi^* = 0$$

$$\partial^\mu(\psi^*\partial_\mu\psi - \psi\partial_\mu\psi^*) = 0$$
$$\partial^\mu J_\mu = 0$$

this is a 4D continuity equation and J^μ is a Noether current.

Expanding the equation yields

$$\frac{\partial}{\partial t}\left(\psi^*\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi^*}{\partial t}\right) - \boldsymbol{\nabla}\cdot(\psi^*\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\psi^*) = 0$$

the second term is proportional to the quantum probability current, but it is missing an i! Moreover, the first term, representing ρ, is **not definite positive**, so it cannot represent the norm of the state $|\psi_t\rangle$!

This proves that it is **impossible** to make QM relativistic using wavefunctions and retaining the probabilistic approach.

Paul Dirac used another approach, he wanted to decrease the order of space derivative (instead of increasing the time derivative order, like K-G): this result was better, but for free particles it yielded a double solution with $E < 0$ states, violating a fundamental QM theorem!
For this reason, he devised the concept of the *Dirac sea*: these $E < 0$ states must all be occupied (because they cannot be observed), and a uniform charge can be factored out of Maxwell's equations with proper boundary conditions; when one of these particles in the sea is excited, the "hole" left behind can be perceived as a "positive electron": he predicted the existence of *antiparticles*.

This interpretation is still classical, but it begins to convey the concept that a finite number of particles is *not* possible for relativistic QM, so the concept of wavefunctions must be left behind in favour of field theories.

3 Symmetries and conservations

To build a theory, it is necessary to start with principles and symmetry: **Lorentz-invariance** to guarantee the conservation of distance between points (this is also useful to define a metric topology!).

Why symmetry? Noether theorem states that for every 4-current that respects a continuity equation

$$\partial_\mu J^\mu = 0$$

there is a conserved Noether charge

$$Q := \int \mathrm{d}^3x J^0, \text{ so that } \frac{\mathrm{d}Q}{\mathrm{d}t} = 0$$

3.1 Lorentz-Poincaré group

The Lorentz invariant is

$$s^2 = t^2 - x_i x^i = t'^2 - x'_i x'^i = x_\mu x^\mu :$$

the Lorentz group is the set of all transformations $x^\mu \to x'^\mu$ over the Minkowski space that conserve s^2; in matrix representation the elements are

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

the invariant is conserved **if and only if**

$$\boxed{g_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = g_{\rho\sigma}.}$$

The covariant quantities transform with the inverse matrix

$$x'_\mu = \Lambda_\nu{}^\mu x'_\nu$$

Observation 1 Since $g_{\mu\nu}$ is a symmetric matrix, this equation encompasses 10 constraints on $\Lambda^{\mu}{}_{\nu}$, leaving 3 degrees of freedom for rotations and 3 for boosts.

Naming convention Consider the determinant of the invariance condition:

$$\det\left(\Lambda^T g \Lambda\right) = \det(g) \iff \det(\Lambda)^2 = 1 \iff \det(\Lambda) = \pm 1$$

Transformations with $\det(\Lambda) = 1$ are called *proper Lorentz transformations.*
Transformations with $\det(\Lambda) = -1$ are called *improper Lorentz transformations.*

Naming convention Consider the (0,0) component of the invariance condition:

$$g_{\mu\nu}\Lambda^{\mu}{}_{0}\Lambda^{\nu}{}_{0} = 1 \iff (\Lambda_{00})^2 - \Lambda^{i}{}_{0} g_{ii} \Lambda^{i}{}_{0} = 1 \iff |\Lambda^{0}{}_{0}|^2 \geq 1$$

Because this term regulates how the transformation relates x^0 and x'^0, transformations with $\Lambda^{0}{}_{0} \geq 1$ are called *orthochronous Lorentz transformations*, while those with $\Lambda^{0}{}_{0} \leq 1$ are called *non-orthochronous Lorentz transformations.*

These 2 divisions split the Lorentz group into 4 parts

- Proper and orthochronous $L_{+}^{\uparrow}$: restricted Lorentz group, sometimes referred to as "Lorentz Group";
- Improper and orthochronous $L_{-}^{\uparrow}$: spacial inversion;
- Proper and non-orthochronous $L_{+}^{\downarrow}$: time inversion;
- Improper and non-orthochronous $L_{-}^{\downarrow}$: total inversion of spacetime.

Note The only subgroup is $L_{+}^{\uparrow}$, as it is the only one that contains the identity matrix $\mathbb{I}$. **Lesson 3, 28/09**

4 Symmetries and Group Theory

The principle of symmetry (invariance) is the guiding light to create a theory: the first and fundamental symmetry is *Lorentz invariance*, because special relativity states that **all physical laws must be the same in every inertial reference frame**; each particle (state) must also observe other symmetries, but those vary case by case.

The language of symmetries is *group theory*, so, in order to proceed, an introduction to groups is needed.

Def. A group is a set G with a binary operation $\cdot$ so that

- $\forall g_1, g_2 \in G$, $g_1 \cdot g_2 \in G$;
- $\exists e \in G$ so that $\forall g \in G$, $g \cdot e = e \cdot g = g$;
- $\forall g \in G$, $\exists g^{-1}$, so that $g^{-1} \cdot g = g \cdot g^{-1} = e$;
- $\forall g_1, g_2, g_3 \in G$, $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3 = g_1 \cdot g_2 \cdot g_3$.

Def. A group is *Abelian* if and only if

$$\forall g_1, g_2 \in G,\ g_1 \cdot g_2 = g_2 \cdot g_1$$

4.1 Zoology

- U meaning "unitary" means $UU^{\dagger} = \mathbb{I}$, for complex valued matrices;
- O meaning "orthogonal" means $OO^T = \mathbb{I}$, for real valued matrices
- S meaning "special" means $\det(S) = 1$.

Examples

- $U(1)$ is the group of phase rotations in $\mathbb{C}$;
- $SU(2)$ is the group of spin;
- $SO(3)$ is the group of proper rotations

Note The number in the name of the group is the dimensionality of the fundamental representation.

5 Lie Groups

The most important groups are those that represent *continuous transformations*, i.e. every element of the group corresponds to a transformation that can be broken down into *infinitesimal transformations*: using parameters α and generators T, the element $g(\alpha) \in G$ is

$$g(\alpha) = \mathbb{I} + i\alpha_a T^a + O(\alpha^2) = ... = \exp\left[i\alpha_a T^a\right]$$

so its generators can be found as

$$T^a = -i \left.\frac{\mathrm{d}g}{\mathrm{d}\alpha_a}\right|_{\alpha=0}$$

it is useful to work with generators as their number is small and they are operators (that can be represented as matrices).

The operation of the groups can be used to construct the operation between the generators, by the BCH formula:

$$g_1 \cdot g_2 \to e^X \cdot e^Y := \exp\left(X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}[X,[X,Y]]\right)$$

where $[\cdot,\cdot]$ is the *Lie commutator*.

Def. The vector space of the generators equipped with a commutator (bilinear, anti-commutative, Jacobi Identity) is called a **Lie Algebra** $\mathfrak{g}$.

Note The most general form of a commutator is

$$[T^a, T^b] = i f^a{}_{bc} T^c$$

5.1 Example: algebra of SO(3)

The rotation matrix about the X axis of an angle ϑ_1 is

$$R_1(\vartheta_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\vartheta_1) & -\sin(\vartheta_1) \\ 0 & \sin(\vartheta_1) & \cos(\vartheta_1) \end{pmatrix}$$

the first generator is then

$$J_1 = -i \left.\frac{\mathrm{d}R_1}{\mathrm{d}\vartheta_1}\right|_{\vartheta_1=0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

The other 2 matrices can be obtained similarly:

$$J_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \qquad J_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The commutation relations turn out to be

$$[J^i, J^j] = i\varepsilon_{ijk} J^k.$$

From the generators, the $J^{\pm}$ operators are constructed.

5.2 Example: algebra of SU(2)

The definition states $SU(2) = \{2\text{x}2 \text{ unitary matrices with} \det(U) = 1\}$: the conditions imply, by the BCH formula:

$$U^\dagger U = \mathbb{I} \to J_i^\dagger = J_i$$
$$\det(U) = \det\left(e^{J_i}\right) = 1 = \exp[\mathrm{Tr}(J_i)] \to \mathrm{Tr}(J_i) = 0$$

the traceless, hermitian matrix vector space has base $\sigma^i i$, the Pauli matrices, which have commutation relations

$$[\sigma^i, \sigma^j] = i\varepsilon_{ijk}\sigma^k$$

6 Group representations

Def. A representation of a group G is a map

$$r : G \to GL(V)$$
$$r : g \to \text{square matrix}$$

so that

$$r(g_1 \cdot g_2) = r(g_1) \cdot r(g_2)$$

where the second $\cdot$ is the matrix multiplication.
$\dim(V) = n$ is the dimension of the representation.

Def. An irreducible representation is a representation in which no subset of $\mathrm{Im}(r)$ is a subgroup of $\mathrm{Im}(r)$.

Note A reducible representation can always be block-diagonalized (?). **Lesson 4, 30/09**

The generators are useful because one can study the entire group just by considering the generators: they are operators uniquely defined by their commutation relations using Lie brackets; the set of generators equipped with their commutation relations is the **Lie algebra** associated with the group.

Once the algebra is known, the choice of the representation yields matrices equivalent to the generators.

As first examples, it is useful to consider $SO(3)$ and $SU(2)$, because they are closely related to the Lorentz group (which is the final purpose of this parenthesis on group theory).

7 Example: angular momentum as a representation

As is known, the algebra of the angular momentum operators J^i is

$$\left[J^i, J^j\right] = i\varepsilon^{ij}{}_k J^k$$

the rank of a group is the maximum number of generators that commute with one another: in this case the rank is 1 and the chosen operator is J^3.
A Casimir operator is an operator that commutes with all generators: in this case it is

$$J^2 = J_i J^i.$$

The physical question is then "how many possible values/eigenstate of the angular momentum are possible?" and it turns out that the answer to this question also answers, in a QFT formalism, "how many particles does the field predict?".
The process to find the possible eigenvalues and eigenstates is known (see QM1 course) and it yields the

fact that for each j value, the number of eigenstates of J^3 is finite and their eigenvalues m must be in the range

$$-j < m < j$$

where j is called the *maximum weight*; so for each j the possible values of J^3 (and so the number of different eigenstates) are $2j+1$. From this process also follows that the only possible values of j are positive integers or semi-integers and that the values of m always differ by an integer (so if j is semi-integer, all the corresponding values of m must also be semi-integer and vice-versa).

This formalism is very powerful because it relates the irreducible representations of the algebra to its vector basis: in a QFT, the dimension of the vector space will correspond to the number of possible physical particles.

This algebra is common between $SU(2)$ and $SO(3)$, with the key difference that for $SO(3)$ only integer values of j are permitted: as seen in intro groups, $SU(2)$ can be represented as a 3-sphere immersed in $\mathbb{R}^4$, while $SO(3)$ is only half 3-sphere in which antipodal points are mapped to the equator (??).

8 Example: reducible representation

The combination of 2 spin $\frac{1}{2}$ particles yields a reducible representation: the vector space is

$$V = \mathbb{C}^2 \otimes \mathbb{C}^2, \quad \dim(V) = 4$$

again, the process of spin combination is treated in the QM1 and QM2 courses and it yields a *triplet* and a *singlet* of states:

$$\text{Triplet (J=1):} \begin{cases} |J=1, M=-1\rangle \\ |J=1, M=0\rangle \\ |J=1, M=-1\rangle \end{cases} \rightarrow \text{Vector states}$$

$$\text{Singlet J=0} : |J=0, M=0\rangle \rightarrow \text{Scalar state}$$

So the space V is subdivided into two different vector spaces which differ in symmetry: the singlet is antisymmetric while the triplet is symmetric; this representation is reducible, because the image of the representation of the algebra is the direct sum of the images of the representations of two different algebras.

9 The Lorentz Group Representation

Only the subgroup $L_+^\uparrow$ is here considered and its formal name is $SO(1,3)$, where $(1,3)$ is the signature of the Minkowski metric $(+,-,-,-)$.

Consider an infinitesimal Lorentz transformation

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad |\omega^\mu{}_\nu| << 1 \, \forall \mu, \nu$$

where $\omega^\mu{}_\nu$ are called *parameters* of the transformation.

Substituting into the invariance condition yields

$$\begin{gathered} g_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = g_{\rho\sigma} \\ g_{\mu\nu}\left(\delta^\mu{}_\rho + \omega^\mu{}_\rho\right)\left(\delta^\nu{}_\sigma + \omega^\nu{}_\sigma\right) = g_{\rho\sigma} \\ g_{\mu\nu}\delta^\mu{}_\rho\delta^\nu{}_\sigma + \omega_{\nu\rho}\delta^\nu{}_\sigma + \omega_{\mu\sigma}\delta^\mu{}_\rho = g_{\rho\sigma} \\ \omega_{\rho\sigma} = -\omega_{\rho\sigma} \end{gathered}$$

so the parameters must be **antisymmetric**. This confirms the fact that each transformation is uniquely determined by 6 quantities. The transformation becomes

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + g^{\mu\rho} g^{\sigma}{}_{\nu} \omega_{\rho\sigma}$$

since $\omega_{\rho\sigma}$ are antisymmetric, the only non-zero part of the product $\omega_{\rho\sigma} g^{\mu\rho} g^{\sigma}{}_{\nu}$ is given by the antisymmetric part of $g^{\mu\rho} g^{\sigma}{}_{\nu}$: anti-symmetrizing it yields

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \frac{1}{2}\omega_{\rho\sigma} \left(g^{\mu\rho} g^{\sigma}{}_{\nu} - g^{\mu\sigma} g^{\rho}{}_{\nu}\right) = \delta^{\mu}{}_{\nu} + i\frac{1}{2}\omega_{\rho\sigma} \left(M^{\rho\sigma}\right)^{\mu}{}_{\nu}$$

confronting with the definition of generators

$$g = \mathbb{I} + i\alpha_a T^a$$

yields that the generators are

$$M^{\rho\sigma} = -i \left(g^{\mu\rho} g^{\sigma}{}_{\nu} - g^{\mu\sigma} g^{\rho}{}_{\nu}\right)$$

so the generic Lorentz transformation can be expressed as

$$\Lambda = \exp\left[\frac{i}{2}\omega_{\rho\sigma} M^{\rho\sigma}\right]$$

and its commutation relations are

$$\left[M^{\lambda\tau}, M^{\rho\sigma}\right] = -i\left(g^{\lambda\sigma} M^{\tau\rho} + g^{\tau\rho} M^{\lambda\sigma} - g^{\lambda\rho} M^{\tau\sigma} - g^{\tau\sigma} M^{\lambda\rho}\right)$$

To make computations easier, the 6 generators are split into 3+3 easier operators: since $SO(3)$ is a subgroup of $SO(1,3)$, there is a way to express half of the Lorentz Group generators in terms of those of $SO(3)$:

$$J_i = \frac{1}{2}\varepsilon_{ijk} M^{jk}$$

which obey the $SO(3)$ algebra

$$[J_i, J_j] = i\varepsilon_{ij}{}^{k} J_k$$

and generate rotations; the boosts are generated by

$$K^i = M^{0i}$$

which obey the commutation relation

$$\left[K^i, K^j\right] = -i\varepsilon^{ij}{}_{k} J^k$$

which means that the algebra of the K's is **not** closed, whereas the algebra of the J's is: this is due to the fact that 3D rotations form a subgroup of $SO(1,3)$, while boosts do not (it is easy to check that the composition of 2 Cartesian boosts is the composition of a Cartesian boost and a rotation). The mixed commutator yields

$$[J_i, K_j] = i\varepsilon_{ijk} J^k$$

so the set of all J's and all K's is the set of the generators of $SO(1,3)$.

Lesson 5, 01/10

10 More on the Lorentz group

The generic Lorentz transformation can also be written in terms of J, K:

$$\Lambda = \exp\left[i\vartheta_i J^i + i\eta_i K^i\right]$$

where ϑ_i are the Euler angles (parameters of the rotation) and η_i are the rapidities (parameters of the boost).

With some work, it is possible to prove that

$$\frac{1}{2}M_{\mu\nu}M^{\mu\nu} = J_i J^i - K_i K^i = \mathbf{J^2} - \mathbf{K^2}$$

and

$$-\frac{1}{4}\varepsilon^{\mu\nu\rho\sigma}M_{\mu\nu}M_{\rho\sigma} = J_i K^i = \mathbf{J}\cdot\mathbf{K}$$

are Casimir operators for the representation: there exist only 2 linear independent Casimir operators and each and any linear combination of these two is valid, but these two are important because they are, respectively, symmetric and antisymmetric (how can they be?? They're scalar!).

Theorem: The algebra of the Lorentz Group can be factorized as the direct sum of two $SU(2)$ algebras

$$\mathfrak{so}(1,3) = \mathfrak{su}(2) \oplus \mathfrak{su}(2).$$

This suggests to find new generators that are suited to this decomposition

$$\mathbf{N} = \frac{1}{2}(\mathbf{J} + i\mathbf{K})$$
$$\mathbf{N}^\dagger = \frac{1}{2}(\mathbf{J} - i\mathbf{K})$$

their algebra is

$$\left[N_i, N_j^\dagger\right] = 0$$
$$[N_i, N_j] = i\varepsilon_{ijk}N^k$$
$$\left[N_i^\dagger, N_j^\dagger\right] = i\varepsilon_{ijk}N^{k\dagger}$$

Observation: These generators are convenient, since the algebra of the N's and that of the $N^\dagger$'s are closed and disjoint.

The Casimir operators are

$$N^2 = N_i N^i \longrightarrow \underset{\text{eigenvalues}}{n(n+1)}$$
$$N^{\dagger 2} = N_i^\dagger N^{i\dagger} \longrightarrow m(m+1),$$

so the irreducible representations of the group will be labeled by the pair (n, m). Since they obey $SU(2)$ algebra, it is clear that n, m will have integer or semi-integer values and that the eigenvalues of $N_3, N_3^\dagger$, s and $s^\dagger$, will respect

$$-n < s < n \qquad -m < s^\dagger < m.$$

(n,m)	Field name	Dim	Example
$(0,0)$	Scalar φ	1	Higgs
$(1/2,0)$	Spinor ψ_L	2	Weyl Spinor
$(0,1/2)$	Spinor ψ_R	2	Weyl Spinor
$(1/2,0)\oplus(0,1/2)$	Dirac Spinor	4	
$(1/2,1/2)$	4-vector field (spin 1)	4	A^μ EM
$(1,0)\oplus(0,1)$	Rank 2 antisymm tensor	6	$F^{\mu\nu}$ EM

To each pair n,m corresponds a name for the field

The generators for each field can be derived: for scalar particles it is easy, consider

$$n=0\rightarrow \mathbf{N}=0=\frac{1}{2}\left(\mathbf{J}+i\mathbf{K}\right)$$
$$m=0\rightarrow \mathbf{N}^\dagger=0=\frac{1}{2}\left(\mathbf{J}-i\mathbf{K}\right)$$

which has only one solution.

$$\mathbf{J}=\mathbf{K}=0$$

For the spinor $(1/2,0)$,

$$n=1/2\rightarrow \mathbf{N}=\frac{1}{2}\sigma$$
$$m=0\rightarrow \mathbf{N}^\dagger=0$$

which implies

$$\begin{cases}(\mathbf{J}+i\mathbf{K})=\sigma\\ \mathbf{J}=i\mathbf{K}\end{cases}\rightarrow\begin{cases}\mathbf{J}=\frac{i}{2}\sigma\\ \mathbf{K}=-\frac{i}{2}\sigma\end{cases}$$

and so the generic transformation can be written as

$$\Lambda_{(1/2,0)}=\exp\left[\frac{i}{2}\left(\vartheta_a\sigma^a-i\eta_a\sigma^a\right)\right].$$

A similar process can be done for $(0,1/2)$ and it yields

$$\Lambda_{(0,1/2)}=\exp\left[\frac{i}{2}\left(\vartheta_a\sigma^a+i\eta_a\sigma^a\right)\right]$$

These two spinorial fields will be united into a single, 4-component field called a *Dirac Field*, (by taking the direct sum of the two components), which has some symmetry properties that these two do not possess singularly and describes **the electron**. Their direct product will yield a vector field that describes **the photon**.

11 Poincaré Group

The QFT must also be invariant over spacetime *translations*: the Poincaré group includes Lorentz transformations *and* translations

$$x^\mu\rightarrow x'^\mu:=\Lambda^\mu{}_\nu x^\nu+a^\mu$$

the group parameters are $6+4=10$. The generators of the 4-translations, similarly to QM, are the 4-momenta

$$P^\mu=(H,P^i);$$

moreover, in QM1 the time evolution operator was found to be

$$U = e^{-iH}$$

which, in hindsight, is an exponential map, in which H is the generator and t is the parameter!

There will be some problems with the fact that the translation group is not compact.

Finding the commutation relations for the algebra is not trivial, since rotations and translations do not commute, even in Minkowski space. Consider two successive Poincaré transformations:

$$x^\mu \to x'^\mu = (\Lambda_{(1)})^\mu{}_\nu x^\nu + a_1^\mu \to x''^\mu = (\Lambda_{(2)})^\mu{}_\rho (\Lambda_{(1)})^\mu{}_\nu x^\nu + (\Lambda_{(2)})^\mu{}_\rho a_1^\rho + a_2^\mu$$

it is clear that they do not commute.
The commutation relations are

$$\begin{aligned}
&[P^\mu, P^\nu] = 0 \\
&[M_{\mu\nu}, P_\rho] = -ig_{\rho\sigma}P_\nu + ig_{\nu\rho}P_\mu
\end{aligned}$$

They are clearer when time and space indexes are expressed separately

$$\begin{aligned}
&\left[P^i, P^j\right] = 0 \quad \left[P^i, H\right] = 0 \quad \left[K^i, P^j\right] = i\delta^{ij}H \\
&\left[J^i, J^j\right] = i\varepsilon^{ij}{}_k J^k \quad \left[K^i, H\right] = iP^i \quad \left[J^i, P^j\right] = i\varepsilon^{ij}{}_k P^k \\
&\left[J^i, H\right] = 0 \quad \left[K^i, K^j\right] = i\varepsilon^{ij}{}_k J^k \quad \left[J^i, K^i\right] = i\varepsilon^{ij}{}_k K^k
\end{aligned}$$

Next, the Casimir operators: the expected number is 2, as there are only 2 commuting generators, a natural choice is

$$P^2 = P_\mu P^\mu \underset{\text{eigenvalue}}{\longrightarrow} m^2,$$

where m is the mass of the particle (as the mass-shell dictates); a not so natural choice is the *Pauli-Lubanski vector* squared:

$$W^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}P_\nu M_{\rho\sigma} \to W^2 = W_\mu W^\mu.$$

A nice relation is

$$P_\mu W^\mu = 0$$

for antisymmetry reasons, and one can show that

$$[W^\mu, P^\nu] = 0$$

Lesson 6, 05/10

12 Massive and massless particles representations

Massive particles
It is known from SR that if a particle is massive, there exist a reference frame in which the particle is at rest, i.e.

$$P^\mu = (m, \mathbf{0})$$

which means that

$$W^0 = 0 \quad W^i = -\frac{1}{2}\varepsilon^{0ijk}M_{jk}$$

recalling the definition of J^i this is

$$W^i = -\frac{1}{2}mJ^i$$

since the particle is at rest, this cannot represent an orbital angular momentum, so it must be intrinsic, i.e. *spin*:

$$W^i = -\frac{1}{2}mS^i$$

and so massive particles are identified by Casimir operators P^2 and $S^2 \propto W^2$ and by the commuting operators S_3 and P^μ: there are, for each value of m, s, $2s+1$ internal degrees of freedom.

Massless particles
If the particle is massless,

$$P_\mu P^\mu = m^2 = 0$$

which means

$$W_\mu W^\mu = 0,$$

and in general

$$P_\mu W^\mu = 0 :$$

from these 3 conditions it can be shown[2] that

$$W^\mu = hP^\mu$$

where $h = \pm 1$ is called **helicity** and represents the projection of the angular momentum on the direction of the momentum

$$h = \frac{\mathbf{J}\cdot\mathbf{P}}{|\mathbf{P}|};$$

in this case, they can only be parallel or anti-parallel: these states are called *polarizations* and, for photons, they correspond to light polarizations.

This framework also allows a sector of the theory in which spin is continuous ($m = 0$, $W^2 < 0$ states are like this, but they have not been observed in nature) and a sector in which some particles, tachyons can move at $v > c$.

The key takeaway from this arguments is that

The irreducible representations of the Poincaré Group are in one-to-one correspondence with the possible types of elementary particles.
The mathematical structure of space-time symmetries determines what kinds of particles can exist in nature.

13 Non-finite representations of groups: field representations

Until this point, all the representations have been finite-dimensional, with generators represented as matrices, but a field is continuous.

Def. A field is a well-behaved function of spacetime

$$\phi_a(x^\mu) = \begin{pmatrix} \phi_1(x^\mu) \\ \phi_2(x^\mu) \\ \dots \end{pmatrix}$$

matrices can transform a field into another with linear combinations of the old components as new components: these transformations are finite-dimensional, while the ones on the coordinates x^μ can also be infinite-dimensional.

[2] moodle?

In QM the allowed transformations were unitary transformations, as they preserve squared norms, which are the observable part of the theory: these transformations are infinite-dimensional (think of the time evolution: its generator is the Hamiltonian H, which is a differential operator).

In parallel to the definition of the angular momentum operator in QM, the Lorentz generator for infinite-dimensional transformation is

$$M_{\mu\nu} = i(x_\mu \partial_\nu - x_\nu \partial_\mu)$$

and so a field transforms like

$$\phi_a(x) \to \phi'_a(x') = M^a{}_b(\Lambda)\phi_b(\Lambda x)$$

where Λ are infinite-dimensional transformations and $M^a{}_b$ are finite-dimensional; the total transformation can be expressed with the exponential map

$$\phi_a(x') = \exp\left[-\frac{i}{2}\omega^{\rho\sigma} M_{\rho\sigma}\right]\phi_b(x),$$

with

$$M_{\mu\nu} = M^{\text{inf}}_{\mu\nu} + M^{\text{fin}}_{\mu\nu} :$$

where the finite part generates the transformations of the field components[3] and the infinite part those of the space-time coordinates. This generators can be expressed together thanks to the BCH formula.

Lesson 7, 06/10

14 Classical Field theory

Before delving into QFT, it is necessary to understand classical field theory.

A classical system is described by the generalized coordinates

$$q_i(t), \text{ with } i = 1, 2...N$$

where N is the (finite) number of degrees of freedom and t is a parameter. From this the Lagrangian is constructed

$$L(q, \dot{q}).$$

A field theory replaces the set of the coordinates$\{q_i\}$ with a *continuous function of space-time*, called a **field**:

$$\phi(t, \mathbf{x}), \ \mathbf{x} \in \mathbb{R}^3$$

the continuous coordinates $\mathbf{x}$ replace the discrete index i and so the degrees of freedom become infinite.

The next step is to construct a Lagrangian density

$$\mathcal{L}(\phi, \partial_\mu \phi)$$

and solve for the equations of motion.

Example: from finite to infinite degrees of freedom.
Suppose a system of N point-like masses interacting with an elastic potential: the Lagrangian will be

$$L = \sum_{k=1}^{N} \left[\frac{1}{2} m_i \dot{x}_i^2 - \frac{1}{2} k(x_i - x_{i+1})^2\right].$$

[3] Q: why are we assuming that the parameters are the same number and are the same?

Consider $m_i = m$ and the distance between them to be constant and equal to ε:

$$L = \sum_{k=1}^{N} \frac{\varepsilon}{2} \left[\frac{m}{\varepsilon} \dot{x}_i^2 - k \frac{(x_i - x_{i+i})^2}{\varepsilon} \right]$$

The limit as ε goes to 0 is

$$\lim_{\varepsilon \to 0} L = \int \mathrm{d}x \frac{1}{2} \left[\mu \dot{\phi}(x)^2 - Y \left(\frac{\partial \phi}{\partial x} \right)^2 \right]$$

where $\frac{m}{\varepsilon} \to \mu$, $k\varepsilon \to Y$ (which is a Young module) and $x_i(t) \to \phi(t, x)$. Applying the least action principle yields the Euler-Lagrange equation

$$\frac{\partial^2 \phi}{\partial x^2} - \frac{\mu}{Y} \frac{\partial^2 \phi}{\partial t^2} = 0.$$

In general, the Lagrangian density is a function of the spacetime coordinates x^μ through the dependency on a field and its derivatives

$$\mathcal{L}(x^\mu) = \mathcal{L}(\phi, \partial_m u \phi)$$

and the Lagrangian is its integral on the physical space

$$L(t) = \int \mathrm{d}^3 x \mathcal{L}(\phi, \partial_\mu \phi);$$

another important quantity is the action, i.e. the integral over spacetime of the density:

$$S = \int \mathrm{d}^4 x \mathcal{L}(\phi, \partial_\mu \phi).$$

All the theories constructed must be *local*, i.e. they must obey causality, so in the Lagrangian density $\mathcal{L}$ will never present terms like

$$\phi(x^\mu)\phi(y^\mu), \text{ with } x^\mu \neq y^\mu$$

which represents an instantaneous interaction between two points of spacetime.

Def. A field theory is local when the dependency con x^μ is

$$\mathcal{L}(x^\mu) = \mathcal{L}(\phi(x^\mu), \partial_\mu \phi(x^\mu))$$

which means that the dynamics of the system do not connect different points in space instantaneously.

14.1 Transformations of fields

Since the theory must be covariant, it is pivotal to understand how fields behave under Poincaré transformations: consider a point in spacetime expressed in coordinates as x^μ and the value of a field at that point $f(x^\mu)$; consider then an infinitesimal transformation $x'^\mu = x^\mu + \delta x^\mu$ so that

$$\Delta f = f'(x'^\mu) - f(x^\mu) = f'(x^\mu + \delta x^\mu) - f(x^\mu) = f'(x^\mu) - f(x^\mu) + \partial_\mu f'(x^\mu) \delta x^\mu + O(\delta^2 x^\mu)$$

at first order, noting that $\partial_\mu f'(x^\mu) = \partial_\mu f(x^\mu) + O(\delta x^\mu)$, this can be written as

$$\Delta f = \delta f + \partial_\mu f(x^\mu) \delta x^\mu.$$

This can be seen as the definition of the operator Δ on fields

$$\Delta = \delta + \delta x^\mu \partial_\mu :$$

under a Poincaré transformation, the variation of a field is made up of the functional variation in the field itself (δ) and the variation in the field given by the transformation of the coordinates (called transport term).

Example: spacetime translations.
Consider the infinitesimal Poincaré transformation

$$x^\mu \to x'^\mu = x^\mu + \varepsilon^\mu$$

by definition, a local field f is invariant under translations, so

$$\Delta f = 0$$

and this means

$$\delta f = -\varepsilon^\mu \partial_\mu f$$

and since $P_\mu = i\partial_\mu$,

$$\delta f = -i\varepsilon^\mu P_\mu f.$$

This means that this functional transformation of the field is generated by the 4-momenta

$$f' = \exp\left[-i\varepsilon^\mu P_\mu\right] f$$

The Lorentz transformations, on the other hand, depend on the representation the field belongs to: the simplest is the scalar representation.

Example: Lorentz transformation of a scalar field.
A scalar field is a field invariant under Poincaré transformations

$$\phi(x^\mu) = \phi'(x'^\mu).$$

As seen before, the translations are easy

$$\phi(x^\mu) \to \phi'(x'^\mu) = \exp\left[-i\varepsilon^\mu P_\mu\right] \phi(x^\mu).$$

Consider now a generic Lorentz transformation

$$x^\mu \to x'^\mu = x^\mu + \varepsilon^{\mu\rho} x_\rho$$

where $\varepsilon^{\mu\rho}$ are the 6 completely antisymmetric parameters of the transformation.
By definition,

$$\Delta\phi = 0 = \delta\phi + \varepsilon^{\mu\rho} x_\rho \partial_\mu \phi$$

and so

$$\delta\phi = -\varepsilon^{\mu\rho} x_\rho \partial_\mu \phi \to \delta = -\varepsilon^{\mu\rho} x_\rho \partial_\mu$$

this operator is non-zero only for the antisymmetric part of $x_\rho \partial_\mu$, so antisymmetrizing yields

$$\delta = -\varepsilon^{\mu\rho} \frac{1}{2} \left(x_\rho \partial_\mu - x_\mu \partial_\rho\right) = -\varepsilon^{\mu\rho} \frac{i}{2} L_{\rho\mu}$$

where $L_{\rho\mu}$ is the generalized angular momentum expressed in its infinite-dimensional form, related to the $M_{\rho\mu}$ generators.
Observation: no spin operator is found, as expected, because the scalar field belongs to the $(0,0)$ representation, and so the only possible angular momentum of the system is orbital.

And so the general Lorentz transformation on a scalar field is

$$\phi'(x'^\mu) = \exp\left[-\frac{i}{2}\varepsilon^{\mu\rho} L_{\rho\mu}\right] \phi(x^\mu)$$

Observation: Any power of a scalar field is a Lorentz scalar (and so all functions of ϕ expressed as power series), the derivative of ϕ is not scalar, while the double derivative is scalar.

Example: Lorentz transformation of a vector field.
Consider the gradient vector field

$$\partial_\mu \phi,$$

where ϕ is a scalar field: what does the operator Δ do to this kind of field?

$$\Delta \partial_\mu \phi = \delta \partial_\mu \phi + \delta x^\nu \partial_\nu \partial_\mu \phi$$

since $\Delta\phi = 0$, this is also

$$\Delta \partial_\mu \phi = [\Delta, \partial_\mu]\, \phi = [\delta, \partial_\mu]\, \phi + [\delta x^\nu \partial_\nu, \partial_\mu]$$

since δ is a linear operator, it commutes with the 4-derivative; expanding δx^ν yields

$$\Delta \partial_\mu \phi = [\varepsilon^{\nu\rho} x_\rho \partial_\nu, \partial_\mu] = \varepsilon^{\nu\rho} x_\rho \partial_\nu (\partial_\mu \phi) - \varepsilon^{\nu\rho} \partial_\mu (x_\rho \partial_\nu \phi)$$

expanding with the product rule

$$\Delta \partial_\mu \phi = \varepsilon^{\nu\rho} x_\rho \partial_\nu \partial_\mu \phi - \varepsilon^{\nu\rho} x_\rho \partial_\mu \partial_\nu \phi - \varepsilon^{\nu\rho} (\partial_\mu x_\rho) \partial_\nu \phi$$

the first two terms are zero, because $\varepsilon^{\nu\rho}\left(x_\rho \partial_\nu \partial_\mu - x_\rho \partial_\mu \partial_\nu\right) = 0$, and $\partial_\mu x_\rho = g_{\mu\rho}$, so

$$\Delta \partial_\mu \phi = -\varepsilon^{\nu\rho} g_{\mu\rho} \partial_\nu \phi = \varepsilon^{\rho\nu} g_{\mu\rho} \partial_\nu \phi.$$

So finally

$$\begin{aligned} \delta(\partial_\mu \phi) + \varepsilon^{\rho\sigma} x_\rho \partial_\sigma \partial_\mu \phi &= \varepsilon^{\rho\nu} g_{\rho\mu} \partial_\nu \phi \\ \delta(\partial_\mu \phi) &= \varepsilon^{\rho\sigma} g_{\mu\rho} g^\nu{}_\sigma (\partial_\nu \phi) - \varepsilon^{\rho\sigma} x_\rho \partial_\sigma (\partial_\mu \phi) \end{aligned}$$

antisymmetrizing both addends with respect to the indices of ε yields

$$\delta(\partial_\mu \phi) = \frac{1}{2} \varepsilon^{\rho\sigma} \left[\underbrace{\left(g_{\mu\rho} g^\nu{}_\sigma - g_{\mu\sigma} g^\nu{}_\rho\right)}_{\text{Independent of } x^\mu} (\partial_\nu \phi) - \underbrace{\left(x_\rho \partial_\sigma - x_\sigma \partial_\rho\right)}_{x^\mu\text{-dependent}} (\partial_\mu \phi) \right].$$

This form is convenient, because it is clear that the transformation of the field has both a x^μ-dependent generator, which is proportional to the generalized angular momentum $L_{\rho\sigma}$, and a generator that does not depend from x^μ, which corresponds to a spin operator $(S_{\rho\sigma})_\mu{}^\nu$: this spin operator is none other than the generator of the finite-dimensional Lorentz transformations $(M_{\rho\sigma})_\mu{}^\nu$ and the orbital angular momentum operator that of the infinite-dimensional transformations.

$$\delta(\partial_\mu \phi) = -\frac{i}{2} \varepsilon^{\rho\sigma} \left(S_{\rho\sigma}\right)_\mu{}^\nu (\partial_\nu \phi) - \frac{i}{2} \varepsilon^{\rho\sigma} L_{\rho\sigma} (\partial_\mu \phi)$$

This example underlines the fact that vector fields can also present internal degrees of freedom, identified as spin, as they are $(n, m) \neq (0, 0)$ irreducible representations of the Poincaré group. **Lesson 8, 07/10**

15 Constructing a Lagrangian

The guiding principle of this process is the Occam's Razor: nature will realize the simplest Lorentz-invariant terms in the Lagrangian.

The fundamental quantity to build a Lagrangian theory is **the action**, defined before as

$$S = \int \mathrm{d}^4x \mathcal{L}(\phi, \partial_\mu \phi) :$$

in classical mechanics the physical trajectory is the one on which the action (as a functional) is minimal; in QM, even if no trajectory exists, this process can be still used in path integral formulation.

As before stated, the Lagrangian density must be *local*, to guarantee causality, *real*, to preserve probability, *Poincaré invariant*, to ensure that the theory is relativistic and must contain *at most a second derivative*, again to guarantee causality.

Other invariances are admitted (and usually present) and are called *internal symmetries*: for example the invariance of a spinor under a global phase shift guarantees **charge conservation**.

16 Least action

The action in natural units is dimensionless, so the Lagrangian density must have units $[\mathrm{m}^{-4}]$=$[\mathrm{kg}^4]$.

Consider an internal symmetry (i.e. a transformation that does not change the spacetime coordinates)[4]:

$$\phi(x) \to \phi'(x') = \phi(x) + \delta\phi(x),$$

the action

$$S = \int \mathrm{d}^4x \mathcal{L}$$

is minimal if

$$\delta S = 0 = \int \mathrm{d}^4x \delta\mathcal{L} = \int \mathrm{d}^4x \left[\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi)\right],$$

where the relation $\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)$ was used; consider

$$\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right) = \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) + \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta\phi \to$$
$$\to \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) = \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right) - \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta\phi :$$

substituting it into the $\delta S = 0$ condition yields

$$0 = \int \mathrm{d}^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right]\delta\phi + \int \mathrm{d}^4x \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right).$$

If the variation of the fields $\delta\phi$ goes to zero at the boundary of the surface enclosing the dominion of integration (i.e. infinity)[5], the second integral is zero by the divergence theorem and the action is stationary if and only if

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\frac{\partial L}{\partial(\partial_\mu\phi)} = 0}$$

which is called the **Euler-Lagrange equation** for fields.

[4] this is equivalent to the small variations of the trajectory in Lagrangian Mechanics.
[5] this is the case for all physical fields.

Note: the E-L equation is invariant under transformations in the form

$$\mathcal{L}' = \mathcal{L} + \partial_\mu \Lambda^\mu$$

where Λ^μ is any 4-vector function: these are called *canonical transformations*. A trivial example of these transformations are simple translations of $\mathcal{L} \to \mathcal{L} + C$.

17 Noether's Theorem

This is one of the pivotal theorems of QFT and all of theoretical physics and it states

To each symmetry of a system corresponds a 4-current satisfying a continuity equation and a conserved quantity, called Noether charge.

Consider a transformation that acts both on the coordinates and a field

$$\begin{cases} x'^\mu = x^\mu + \delta x^\mu \\ \phi'(x') = \phi(x) + \Delta\phi(x) \end{cases}$$

which does not change the action S[6]. Staring with ϕ as a solution to the E-L equation, the action is invariant under the transformation if and only if

$$\Delta S = 0 = \int \mathrm{d}^4x \Delta\mathcal{L} + \int \delta(\mathrm{d}^4x)\mathcal{L}$$

where

$$\Delta\mathcal{L} = \underbrace{\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\partial_\mu\delta\phi}_{=\delta\mathcal{L}} + \partial_\mu\mathcal{L}\delta x^\mu = \underbrace{\left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\right)}_{=0 \text{ E-L}}\delta\phi + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi\right) + (\partial_\mu\mathcal{L})\delta x^\mu.$$

In this case, since the transformation acts on the coordinates, the variation of the integration measure must be considered:

$$\delta(\mathrm{d}^4x) = \mathrm{d}^4x' - \mathrm{d}^4x$$

using the Jacobian of the coordinate transformation

$$\mathrm{d}^4x' = |\det(J(x))|\mathrm{d}^4x = \left|\det\frac{\partial x'^\mu}{\partial x^\nu}\right|\mathrm{d}^4x$$

$$\delta(\mathrm{d}^4x) = \mathrm{d}^4x' - \mathrm{d}^4x \approx \partial_\mu\delta x^\mu \mathrm{d}^4x,$$

since

$$\frac{\partial x'^\mu}{\partial x^\mu} = \frac{\partial(x^\mu + \delta x^\mu)}{\partial x^\nu} = g^\mu{}_\nu + \partial_\nu\delta x^\mu,$$

the property $\det A = e^{\mathrm{Tr}(\ln A)}$ yields

$$\det J(x) \underset{\delta x^\mu \to 0}{\approx} \exp\left[\mathrm{Tr}\,\partial_\nu\delta x^\mu\right] \sim 1 + \partial_\mu\delta x^\mu.$$

From this follows that the variation of the measure is

$$\delta(\mathrm{d}^4x) \approx \partial_\mu\delta x^\mu \mathrm{d}^4x.$$

[6]This is the definition of symmetry

So the action is stationary if and only if

$$0 = \Delta S = \int \mathrm{d}^4x \left[\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi \right) + \underbrace{\partial_\mu \mathcal{L} \delta x^\mu + \mathcal{L} \partial_\mu \delta x^\mu}_{=\partial_\mu(\mathcal{L}\delta x^\mu)} \right] =$$
$$= \int \mathrm{d}^4x \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi + \mathcal{L}\delta x^\mu \right) = 0$$

this happens if and only if

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi + \mathcal{L}\delta x^\mu \right) = 0.$$

The function in parenthesis is *the Noether current*

$$\boxed{J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi + \mathcal{L}\delta x^\mu}$$

and the found relation is its continuity equation, as

$$\partial_\mu J^\mu = \partial_0 J^0 + \partial_i J^i = \frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0,$$

where $J^0 := \rho$, the density associated with the 3-current $\mathbf{j}$.

Integrating the continuity equation yields

$$0 = \int \mathrm{d}^3x \partial_\mu J^\mu = \int \mathrm{d}^3x \partial_0 J^0 + \underbrace{\int \mathrm{d}^3x \partial_i J^i}_{=0 \text{ by div thm}} = \int d^3x \frac{\partial J^0}{\partial t} = \frac{\mathrm{d}}{\mathrm{d}t} \int \mathrm{d}^3x J^0 := \frac{\mathrm{d}Q}{\mathrm{d}t} = 0$$

so the Noether charge which is conserved is

$$\boxed{Q = \int d^3x J^0(x^\mu).}$$

It is useful to rewrite the Noether current in terms of $\Delta\phi$: recalling that

$$\delta\phi = \Delta\phi - \partial_\nu \phi \delta x^\nu$$

the current becomes

$$\boxed{J^\mu = \underbrace{\left(\mathcal{L} g^\mu{}_\nu - \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi \right)}_{:=-T^\mu{}_\nu} \delta x^\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \Delta\phi}$$

where $T^\mu{}_\nu$ is called the *stress energy tensor*; the sign is a convention that ensures $T^0{}_0 \geq 0$.

Lesson 9, 08/10

Observation The action is invariant under canonical transformations (because the E-L equation is), but the Lagrangian density is not: this produces an ambiguity in the Noether current J^μ, which is invariant under transformations in the form

$$J^\mu \to J'^\mu = J^\mu + \partial_\rho t^{\rho\mu}, \text{ with } t^{\rho\mu} = -t^{\mu\rho}$$

17.1 Transformation parameters

It is convenient to drop the dependency on the parameters of the transformation:

$$\delta x^\rho = \frac{\partial \delta x^\rho}{\partial \omega^A}\delta\omega^A$$
$$\delta\phi = \frac{\partial \delta\phi}{\partial \omega^A}\delta\omega^A$$

where ω^A are the parameters of the symmetry and A is a set of indices: in general the Noether current is in the form

$$J'^\mu = J^\mu{}_A\,\delta\omega^A$$

and, because the continuity equation holds $\forall\omega^A$,

$$\partial_\mu J^\mu{}_A\,\delta\omega^A = 0 \iff \partial_\mu J^\mu{}_A = 0.$$

This form underlines the fact that the number of charges also depends on the indices of the parameters:

$$Q_A = \int \mathrm{d}^3x J^0_A(x^\mu) \to \frac{\mathrm{d}Q_A}{\mathrm{d}t} = 0.$$

17.2 Examples

- **Internal symmetries**
  Consider a field ϕ and its symmetry

  $$\phi \to \phi + \delta\phi, \quad \delta x^\mu = 0$$

  The Noether current is

  $$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi$$

  and the charges are

  $$Q = \int \mathrm{d}^3x J^0$$

- **Space-time translations**
  Consider the transformation

  $$x^\mu \to x'^\mu = x^\mu + \varepsilon^\mu, \text{ with } \varepsilon^\mu << 1$$

  Since ϕ is a local field, $\Delta\phi = 0$ and

  $$\delta\phi = -\varepsilon^\mu\partial_\mu\phi$$

  The current is

  $$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^\alpha}\left(-\varepsilon^\nu\partial_\nu\phi^\alpha\right) + \mathcal{L}\varepsilon^\mu = \varepsilon^\nu\left[\mathcal{L}g^\mu{}_\nu - \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^\alpha}\partial_\nu\phi^\alpha\right] = -\varepsilon^\nu T^\mu{}_\nu$$

  as seen before, the continuity equation must hold $\forall\varepsilon^\nu$, so

  $$\partial_\mu T^\mu{}_\nu = 0.$$

  The Noether charges are

  $$Q^\mu := P^\mu = \int \mathrm{d}^3x T^0{}_\mu$$

  The 0-component is the total energy of the system, i.e. the Hamiltonian

  $$P^0 = H = \int \mathrm{d}^3x T^0{}_0 = \int \mathrm{d}^3x\mathcal{H}$$

where $\mathcal{H}$ is the Hamiltonian density.
This quantity can also be defined using the conjugated "coordinate" to ϕ

$$\pi(x^\mu) = \frac{\partial \mathcal{L}}{\partial \partial_0 \phi} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

then

$$\mathcal{H} = \pi\dot{\phi} - \mathcal{L}$$

18 The action of a scalar field

The Lagrangian density of a scalar field is

$$\boxed{\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi(x^\mu))}$$

which includes a second derivative (since this is Lorentz-invariant) as a kinetic term and a potential term that depends only on ϕ.

The dimension of $\mathcal{L}$ turn out to be $[\mathcal{L}] = m^{-4} = kg^4$, so that of the field is $[\phi] = m^{-1} = kg$.

Using these dimensionality considerations, the simplest Lagrangian is the **Lagrangian of a non-interacting system**

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

From this, the E-L equation is the equation of motion for the field

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} &= 0\\ -m^2\phi - \partial_\mu(\partial^\mu\phi) &= 0\\ (\partial_\mu\partial^\mu + m^2)\phi(x^\mu) &= 0\end{aligned}$$

this is the K-G equation! This can also be interpreted as an eigenvalue equation for the Casimir operator $\Box = -P^2$, which implies

$$(-P^2 + m^2)\phi = 0 \iff P^2 = m^2 :$$

this means that the dimensional parameter m in $\mathcal{L}$ is indeed the mass associated with the field.

18.1 Symmetries: translations

The Noether current for translations, the stress-energy tensor, in this case comes out to be

$$J_{\mu\nu} = T_{\mu\nu} = -g^{\mu\nu}\mathcal{L} + \partial_\mu\phi\partial_\nu\phi$$

and the charges, the momenta, are

$$P_\mu = \int \mathrm{d}^3x J_{\mu 0} = \int \mathrm{d}^3x(-g_{\mu 0}\mathcal{L} + \partial_\mu\phi\partial_0\phi).$$

The T_{00} component is

$$T_{00} = -\mathcal{L} + \partial_0\phi\partial_0\phi = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}\partial_i\phi\partial^i\phi + \frac{1}{2}m^2\phi^2 := \mathcal{H}$$

18.2 Symmetries: Lorentz invariance

Here $\delta x^\mu = \varepsilon^{\mu\nu} x_\nu$ and, by definition, $\Delta\phi = 0$, so $\delta\phi = -\varepsilon^{\mu\nu} x_\nu \partial_\mu \phi$: the Noether current is

$$J^\mu = \left(\mathcal{L} g^\mu{}_\nu - \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi \right) \delta x^\nu,$$

some boring calculations yield

$$J^\mu = -\varepsilon^{\nu\rho} \left(\partial^\mu \phi \partial_\nu \phi x_\rho - \mathcal{L} g^\mu{}_\nu x_\rho \right);$$

before eliminating the parameters $\varepsilon^{\nu\rho}$, it is necessary to antisymmetrize the parenthesis, the result is

$$J^\mu = -\frac{1}{2} \varepsilon^{\nu\rho} \underbrace{\left[\left(\partial^\mu \phi \partial_\nu \phi - \mathcal{L} g^\mu{}_\nu \right) x_\rho - \left(\partial^\mu \phi \partial_\rho \phi - \mathcal{L} g^\mu{}_\rho \right) \right]}_{:= J^\mu{}_{\nu\rho}}.$$

$$\partial_\mu J^\mu = 0 \iff \partial_\mu J^\mu{}_{\nu\rho} = 0 \iff \partial_\mu J^{\mu\nu\rho} = 0$$

So the parameter-independent Noether currents are $J^{\mu\nu\rho}$: they can be written in terms of the energy-momentum tensor

$$J^{\mu\nu\rho} = T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu.$$

It can be shown that the corresponding charges are the generators of Lorentz transformations in its infinite representation (?):

$$Q^{\nu\rho} = \int \mathrm{d}^3 x J^{0\nu\rho} = M^{\nu\rho} \to \frac{\mathrm{d}Q^{\nu\rho}}{\mathrm{d}t} = 0$$

so the angular momenta are conserved, since

$$L^i = \frac{1}{2} \varepsilon^{ijk} M_{jk},$$

and so are the boost generators

$$K^i = M^{0i}.$$

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18.3 Charged scalar field

For the simple scalar field considered earlier, no internal symmetries can be found (?): to generalize the concept, one must consider a complex field, which has multiple components while retaining the fact that it is scalar: consider two independent real scalar fields ϕ_1, ϕ_2 and their linear combinations

$$\phi(x^\mu) = \frac{1}{\sqrt{2}} (\phi_1 + i\phi_2)$$
$$\phi^\dagger(x^\mu) = \frac{1}{\sqrt{2}} (\phi_1 - i\phi_2)$$

The simplest Lagrangian using this kind of field presents the simplest bilinear Lorentz invariance:

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^\dagger - m^2 \phi^\dagger \phi$$

The Euler-Lagrange equations turn out to be

$$\begin{cases} \left(\Box + m^2 \right) \phi(x^\mu) = 0 \\ \left(\Box + m^2 \right) \phi^\dagger(x^\mu) = 0 \end{cases}.$$

Consider the internal transformation that gives an arbitrary phase shift to the field

$$\phi \to \phi' = e^{i\alpha}\phi \to \phi' = \phi + i\alpha\phi$$
$$\phi \to \phi^{\dagger\prime} = e^{i\alpha}\phi \to \phi^{\dagger\prime} = \phi^\dagger + i\alpha\phi^\dagger$$

this is a symmetry with respect to the group $U(1)$.

The Noether current is

$$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^\dagger}\delta\phi^\dagger,$$

expanding the terms and factorizing the parameter α yields

$$J^\mu = \partial^\mu\phi^\dagger(i\phi) + \partial^\mu\phi(-i\phi^\dagger) = i\left(\phi^\dagger\partial_\mu\phi - \phi\partial_\mu\phi^\dagger\right).$$

Noether's theorem states that the quantity

$$Q = \int \mathrm{d}^3x J^0 = i\int \mathrm{d}^3x \left(\phi^\dagger\dot{\phi} - \phi\dot{\phi}^\dagger\right)$$

is conserved: in classical field theory, this does not seem like much, but with canonical quantization it will be clear that this Q is **the total charge of the system**.

19 Spinor Field

In this section the spinor field will be defined and its invariances will be studied.

19.1 Spinor representations of the Lorentz Group

As hinted before, there are 2 irreducible representations associated with spinor fields, both related to the $\mathfrak{su}(2)$ algebra:

$$\left(\frac{1}{2}, 0\right) \to \psi_L(x^\mu) = \begin{pmatrix}\psi_L^1(x^\mu)\\ \psi_L^2(x\mu)\end{pmatrix}$$

$$\left(0, \frac{1}{2}\right) \to \psi_R(x^\mu) = \begin{pmatrix}\psi_L^1(x^\mu)\\ \psi_L^2(x^\mu)\end{pmatrix}$$

Consider now the Lorentz transformations Λ_L and Λ_R, which are 2×2 matrices and are expected to be generated by the angular momenta J^i and boosts K^i operators, with the caveat, given by Wigner's theorem, that, since they do not form a compact group, the K's do not admit unitary representation:

$$J^i = \frac{1}{2}\sigma^i \to \exp\left[\frac{i}{2}\omega_i\sigma^i\right]$$
$$K^i = \mp i\frac{\sigma^i}{2} \to \exp\left[\mp\frac{1}{2}\eta_i\sigma^i\right]$$

Where the $\mp$ sign distinguishes

$$K_L^i = -i\frac{\sigma^i}{2},\ K_R^i = +i\frac{\sigma^i}{2}.$$

So the Lorentz transformations are

$$\Lambda_L = \exp\left[\frac{1}{2}\sigma^i\left(i\omega_i - \eta_i\right)\right]$$
$$\Lambda_R = \exp\left[\frac{1}{2}\sigma^i\left(i\omega_i + \eta_i\right)\right].$$

These transformation are related:

$$\Lambda_L^{-1} = \Lambda_R^\dagger,\ \sigma^2\Lambda_L\sigma^2,\ \Lambda_L^T\sigma^2\Lambda_L = \sigma^2,\ \Lambda_L^T = \sigma^2\Lambda_L^{-1}\sigma_2$$

and the relations hold switching R and L; these properties produce an interesting result

$$\sigma^2\psi_L^* \underset{\text{L.T.}}{\to} \sigma^2\Lambda_L^*\psi_L^* = \sigma^2\Lambda_L^*\sigma^2\sigma^2\psi_L^* = \Lambda_R\sigma^2\psi_L^*$$

which means that $\sigma^2\psi_L^*$ transforms like an element of the $\left(0, \frac{1}{2}\right)$ representation.

19.2 Invariants built from spinors

From QM1 spin composition, it is known that

$$\left(\frac{1}{2}, 0\right) \otimes \left(0, \frac{1}{2}\right) = (0,0) \oplus (1,0) :$$

consider two left spinors ψ_L, χ_L and the object

$$\chi^T\sigma^2\psi_L,$$

how does it transform?

$$\chi^T\sigma^2\psi_L \underset{\text{L.T.}}{\to} \chi_L^T\Lambda_L^T\sigma^2\Lambda_L\psi_L = \chi^T\sigma^2\psi_L$$

This combination of two spinors of the left representation is a scalar; consider the special case $\psi_L = \chi_L$:

$$\psi_L^T\sigma^2\psi_L = \begin{pmatrix}\psi_L^1 & \psi_L^2\end{pmatrix}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\begin{pmatrix}\psi_L^1\\ \psi_L^2\end{pmatrix} = -i\psi_L^1\psi_L^2 + i\psi_L^2\psi_L^1$$

in order for this expression to be non-zero, the components of the spinors must be anticommuting numbers, called *Grassman Numbers*.

Considering $\chi_L = \sigma^2\psi_R^*$ and multiplying by i, yields

$$i(\sigma^2\psi_R^*)^T\sigma^2\psi_L = -i\psi_R^\dagger\psi_L$$

Equivalently, one can prove that another invariant is

$$\psi_L^\dagger\psi_R.$$

Another way to determine invariants uses the 4D extension of the Pauli matrices

$$\sigma^\mu = (\mathbb{I}, \sigma^i),$$
$$\bar{\sigma}^\mu = (\mathbb{I}, -\sigma^i)$$

to build 4-vectors like

$$\psi_L^\dagger\bar{\sigma}^\mu\psi_L = (\psi_L^\dagger\psi_L, -\psi_L^\dagger\sigma^i\psi_L)$$
$$\psi_R^\dagger\sigma^\mu\psi_R = (\psi_R^\dagger\psi_R, \psi_R^\dagger\sigma^i\psi_R) :$$

one can use derivatives contracted with the index to create invariants:

$$\psi_L^\dagger\bar{\sigma}^\mu\partial_\mu\psi_L + \psi_R^\dagger\sigma^\mu\partial_\mu\psi_R$$

Note: All these invariant are linear combinations of the two representations: this hints at the fact that they should not be treated separately.

19.3 Dirac Spinor

There is a problem with the 2 spinor representations, since the J's are pseudo-vectors and the K's are not, a parity transformation acts differently on the generators and on the Casimir operators: it interchanges the operators $N^2, N^{\dagger 2}$ and so the eigenvalues that label the representations

$$(n,m) \underset{\text{Parity}}{\Longleftrightarrow} (m,n).$$

This means that one of these representations is not enough to describe a spin 1/2 field which is invariant under parity transformations (because reality is!).

The needed field is the **Dirac Spinor**, defined in the *Weyl representation* as

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_L^1 \\ \psi_L^2 \\ \psi_R^1 \\ \psi_R^2 \end{pmatrix}$$

19.4 Expanding the group

To unite the two spinors, a broader group needs to be used: recalling the definition of the sectors of the Lorentz Group, the needed group is the set of proper and improper orthochronous Lorentz transformations

$$L^{\uparrow} = L_+^{\uparrow} \cup L_-^{\uparrow},$$

which contains parity operators. In this group, the sum of the two representations of the restricted Lorentz group is *an irreducible representation*: the generators are the 4 Dirac γ^μ matrices, which in the Weyl representations are

$$\gamma^\mu = \begin{pmatrix} \mathbb{O} & \sigma^\mu \\ \bar{\sigma}^\mu & \mathbb{O} \end{pmatrix} \longrightarrow \gamma_0 = \begin{pmatrix} \mathbb{O} & \mathbb{I} \\ \mathbb{I} & \mathbb{O} \end{pmatrix}, \gamma^i = \begin{pmatrix} \mathbb{O} & \sigma^i \\ \sigma^i & \mathbb{O} \end{pmatrix}$$

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The role of the improper transformations of $L_-^{\uparrow}$ is to *connect* the two representations: the representation associated with the Dirac spinor is the $(1/2, 0) \oplus (0, 1/2)$.

19.5 Weyl representation

In the Weyl representation, the γ^0 matrix is the operator that exchanges ψ_L and ψ_R, while the γ^i are the 4×4 extension of the Pauli matrices.

They have an important property

$$(\gamma^0)^2 = \mathbb{I}, \ \left(\gamma^i\right)^2 = -\mathbb{I}$$

and their algebra satisfies

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{I}.$$

Note: This is the first anti-commutation relation encountered in this course.

The gamma matrices form a structure called a **Clifford Algebra** and are not Hermitian matrices, since

$$\left(\gamma^0\right)^\dagger = \gamma^0, \ \left(\gamma^i\right)^\dagger = -\gamma^i \longrightarrow (\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$$

There is another matrix, called *chirality matrix*

$$\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3 \longrightarrow (\gamma^5)^2 = \mathbb{I}, \ \{\gamma_5, \gamma^\mu\} = 0, \ \gamma_5^\dagger = \gamma_5$$

19.6 Action and Lagrangian for Spinor Fields

As seen before, the kinetic terms contain derivatives and an invariant is the sum of the two separate invariants found before

$$\mathcal{L}^K = \underbrace{i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L}_{\text{kinetic L}} + \underbrace{i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R}_{\text{kinetic R}} = i\bar{\psi}\gamma^\mu \partial_\mu \psi$$

where $\bar{\psi} = \psi^\dagger \gamma^0$: to check if this is an invariant, the kinetic term explicitly is

$$\mathcal{L} = i \begin{pmatrix} \psi_L^\dagger & \psi_R^\dagger \end{pmatrix} \begin{pmatrix} \mathbb{O} & \mathbb{I} \\ \mathbb{I} & \mathbb{O} \end{pmatrix} \begin{pmatrix} \mathbb{O} & \sigma^\mu \\ \bar{\sigma}^\mu & \mathbb{O} \end{pmatrix} \partial_\mu \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} =$$

$$= i \begin{pmatrix} \psi_L^\dagger & \psi_R^\dagger \end{pmatrix} \begin{pmatrix} \bar{\sigma}^\mu & \mathbb{O} \\ \mathbb{O} & \sigma^\mu \end{pmatrix} \begin{pmatrix} \partial_\mu \psi_L \\ \partial_\mu \psi_R \end{pmatrix} = i \left(\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L + \psi_R^\dagger \sigma^\mu \partial_\mu \psi_R \right)$$

The mass term is simpler and is

$$\mathcal{L}^m = -m\bar{\psi}\psi = -m \left(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L \right)$$

Finally, the Dirac Lagrangian is

$$\boxed{\mathcal{L}_D = \bar{\psi} \left(i\gamma^\mu \partial_\mu - m \right) \psi}$$

with a quick dimensional check, m is indeed the mass associated to the field, while the dimension of ψ is $L^{-3/2}$.

Note: Another possible mass term is

$$\mathcal{L}_5^m = -im\bar{\psi}\gamma^5\psi :$$

this will be explained later.

19.7 Dirac equation and Noether charges

Applying the E-L equation with respect to $\bar{\psi}, \psi$ yields the equations of motion

$$\boxed{\left(i\gamma^\mu \partial_\mu - m \right) \psi = 0}$$

$$\boxed{i\partial_\mu \bar{\psi} \gamma^\mu + m\bar{\psi} = 0.}$$

The first of these equations is the **Dirac equation**.
By construction, the Lagrangian is invariant under phase transformations ($U(1)$) of the fields:

$$\begin{cases} \psi \to \psi' = e^{i\alpha}\psi \to \delta\psi = i\alpha\psi \\ \bar{\psi} \to \bar{\psi}' = e^{-i\alpha}\bar{\psi} \to \delta\bar{\psi} = -i\alpha\bar{\psi} \end{cases}$$

Since this is an internal symmetry, the Noether current is

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \psi} \delta\psi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \bar{\psi}} \delta\bar{\psi} = -\alpha\bar{\psi}\gamma^\mu\psi,$$

getting rid of the parameter the current turns out to be

$$J^\mu = \bar{\psi}\gamma^\mu\psi$$

which is called the *vector current* and will turn out to be *exactly* the electric current after quantizing this field. The Noether charge is

$$Q = \int \mathrm{d}x^3 J^0 = \int \mathrm{d}^3 x \bar{\psi}\gamma^0\psi = \int \mathrm{d}^3 x \psi^\dagger \gamma^0 \gamma^0 \psi = \int \mathrm{d}^3 x \psi^\dagger \psi = \int \mathrm{d}^3 x \left(\psi_L^\dagger \psi_L + \psi_R^\dagger \psi_R \right)$$

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The Dirac Kinetic Lagrangian admits another continuous internal symmetry, the *chiral* phase transformation

$$\psi \to \psi' = e^{i\beta\gamma_5}\psi \to \delta\psi = i\beta\gamma_5$$

and its Noether current turns out to be

$$J_5^\mu = \bar{\psi}\gamma^\mu\gamma_5\psi$$

and is called the *axial current*. The associated Noether charge is

$$Q_5 = \int \mathrm{d}^3x\bar{\psi}\gamma^0\gamma_5\psi = \int \mathrm{d}^3x \left(\psi_L^\dagger\psi_L - \psi_R^\dagger\psi_R\right)$$

Chirality is important for QFT, because it can only be "detected" by weak interactions: the Weyl representation is convenient to distinguish the left and right spinor, while the other, the **Dirac** representation, is useful to distinguish particles from antiparticles.

19.8 Properties of the Dirac Equation

Notation: Given a 4-vector a^μ, it is sometimes useful to use

$$\not{a} := \gamma^\mu a_\mu,$$

even for the 4-gradient:

$$\not{\partial} = \gamma^\mu\partial_\mu.$$

So the Dirac equation can be written as

$$(i\not{\partial} - m)\psi = 0.$$

19.8.1 Covariance

Consider a Lorentz transformation of the spinor:

$$\psi(x^\mu) \to \psi'(x'^\mu) = S(\Lambda)\psi(x^\mu)$$

where

$$S(\Lambda) = e^{-i/2\,\omega_{\rho\sigma}M^{\rho\sigma}} :$$

how does the Dirac equation transform?

$$(i\gamma^\mu\partial'_\mu - m)\psi'(x')$$

the transformation of the derivative is

$$\partial'_\mu = \underbrace{\Lambda_\mu{}^\nu}_{:=(\Lambda^\mu{}_\nu)^{-1}}\partial_\nu$$

and the γ^μ are independent of x^μ: the equation becomes

$$(i\gamma^\mu\Lambda_\mu{}^\nu\partial_\nu - m)S(\Lambda)\psi(x^\mu) = 0$$

Applying $S^{-1}(\Lambda)$ yields

$$\left(S^{-1}(\Lambda)\gamma^\mu S(\Lambda)\Lambda_\mu{}^\nu\partial_\nu - m\right)\psi(x^\mu) = 0$$

Note: The $S(\Lambda)$ operator and the γ^μ operate in the same vector space (that of Dirac spinors), while Λ and ∂ operate on the spacetime coordinates: this implies that $S(\Lambda)$ does not commute with γ^μ, but it does with Λ and ∂.

The equation is covariant if and only if

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \gamma^\rho\Lambda^\mu{}_\rho :$$

this means that the γ^μ *transform as a Lorentz vector.*

19.8.2 Generators of the Lorentz Group

Recalling

$$S(\Lambda) = \exp\left[-\frac{i}{2}\omega_{\rho\sigma}M^{\rho\sigma}\right],\ M^{0i} = -\frac{i}{2}\begin{pmatrix}\sigma^i & \mathbb{O}\\ \mathbb{O} & -\sigma^i\end{pmatrix},\ M^{ij} = \frac{1}{2}\varepsilon^{ijk}\begin{pmatrix}\sigma_k & \mathbb{O}\\ \mathbb{O} & \sigma_k\end{pmatrix},$$

one can define

$$\sigma^{\mu\nu} = \frac{i}{2}\left[\gamma^\mu, \gamma^\nu\right]$$

so that the Lorentz generators, in the finite representation of the Lorentz algebra, can be expressed as

$$M^{\mu\nu} = \frac{1}{2}\sigma^{\mu\nu}.$$

Note This representation solves the problem (encountered with the left and right spinors) of the boosts with different signs, since the generators of the boosts are

$$K^i = \frac{1}{2}\begin{pmatrix}\sigma^i & \mathbb{O}\\ \mathbb{O} & -\sigma^i\end{pmatrix},$$

while the rotations are generated by

$$M^{ij} = \frac{1}{2}\varepsilon^{ijk}\begin{pmatrix}\sigma_k & \mathbb{O}\\ \mathbb{O} & \sigma_k\end{pmatrix}:$$

the rotation generators will be the same in the Dirac representation, while the boosts' will be different.

Using this formalism, it is easy to show that

$$\psi^\dagger\psi$$

is not invariant, since

$$S(\Lambda)^\dagger S(\Lambda) \neq \mathbb{I},$$

by Wigner's theorem (the boosts are not compact). And also that

$$\bar{\psi}\psi = \psi^\dagger\gamma^0\psi$$

is invariant: it can be proven that

$$\gamma^0(\sigma^{\mu\nu})^\dagger\gamma^0 = \sigma^{\mu\nu};$$

consider then

$$\gamma^0 S^\dagger(\Lambda)\gamma^0 = \gamma^0\left(\mathbb{I} + \frac{i}{4}\omega_{\mu\nu}(\sigma^{\mu\nu})^\dagger\right)\gamma^0 = \mathbb{I} + \frac{i}{4}\omega_{\mu\nu}\gamma^0(\sigma^{\mu\nu})^\dagger\gamma^0 = \mathbb{I} + \frac{i}{4}\omega_{\mu\nu}\sigma^{\mu\nu} = S^{-1}(\Lambda)$$

and so

$$\bar{\psi}\psi = \psi^\dagger\gamma^0\psi \longrightarrow \bar{\psi}'\psi' = \left(S(\Lambda)\psi\right)^\dagger\gamma^0 S(\Lambda)\psi = \psi^\dagger S^\dagger\gamma^0 S\psi$$

inserting $\gamma^0\gamma^0 = \mathbb{I}$ yields

$$\bar{\psi}'\psi' = \psi^\dagger\gamma^0\underbrace{\gamma^0 S^\dagger\gamma^0}_{=S^{-1}}S\psi = \psi^\dagger\gamma^0 S^{-1}S\psi = \bar{\psi}\psi$$

thus proving the invariance.

It can be similarly proven that $\bar{\psi}\gamma^\mu\psi$ is a Lorentz vector:

$$\bar{\psi}\gamma^\mu\psi \longrightarrow \bar{\psi}S^{-1}\gamma^\mu S\psi = \Lambda^\mu{}_\nu\bar{\psi}\gamma^\nu\psi$$

19.8.3 Klein-Gordon strikes back

Another way to write the equation is by applying $-i\gamma^\mu\partial_\mu - m$ to the left:

$$(-i\gamma^\mu\partial_\mu - m)(i\gamma^\nu\partial_\nu - m)\psi = 0$$
$$(\gamma^\mu\gamma^\nu\partial_\mu\partial_\nu + m^2)\psi = 0$$

the $\gamma^\mu\gamma^\nu$ needs to be symmetrized and yields the anticommutator

$$(\frac{1}{2}\underbrace{\{\gamma^\mu,\gamma^\nu\}}_{=2g^{\mu\nu}}\partial_\mu\partial_\nu + m^2)\psi = 0$$

and so the Klein Gordon equation pops out

$$\left(\Box + m^2\right)\psi = 0$$

which means that **each component of the Dirac spinor is a scalar field**, because its equation of motion is the K-G equation.

19.8.4 Parity invariance

The reason Dirac spinors were introduced was to construct a parity-invariant theory: how do these spinors behave under parity transformations?

Consider a parity transformation

$$x^\mu \to x'^\mu = (x^0, -\mathbf{x})$$

which acts on the field as

$$\psi(x^\mu) \to \psi'(x'^\mu) = S_p\psi(x^\mu).$$

The Dirac equation transforms like

$$(i\gamma^0\partial_0 - i\gamma^i\partial_i - m)S_p\psi(x^\mu) = 0$$

applying S_p^{-1} to the left

$$(iS_p^{-1}\gamma^0S_p\partial_0 - iS_p^{-1}\gamma^iS_p\partial_i - m)\psi(x^\mu)$$

This is invariant if and only if

$$S_p^{-1}\gamma^0S_p = \gamma^0,\ S_p^{-1}\gamma^iS_p = \gamma^i \iff S_p = \eta_p\gamma^0$$

where η_p is a complex constant called *intrinsic parity*.

Since the quantity $\psi^\dagger\psi$ is invariant under parity (because the parity transformation is represented as a unitary matrix), the intrinsic parity η_p must be so that

$$|\eta_p|^2 = 1 \iff \eta_p = \pm 1$$

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19.8.5 Recap of Covariants

- $\bar{\psi}\psi$ is a scalar;
- $\bar{\psi}\gamma_5\psi$, since γ_5 anticommutes with γ^μ, it commutes with S and it is a scalar, but it changes sign under parity ($\gamma^0, \gamma_5 = 0$), so this is a *pseudo*scalar;
- $\bar{\psi}\gamma^\mu\psi$ is a Lorentz vector: it is the Noether current associated to the phase invariance;
- $\bar{\psi}\gamma^\mu\gamma_5\psi$ is a Lorentz pseudovector: it is the Noether current associated to the chiral phase invariance;
- $\bar{\psi}\sigma^{\mu\nu}\psi$ is a twice-covariant antisymmetric Lorentz tensor.

The 5 quantities between $\bar{\psi}$ and ψ, which are

$$\mathbb{I},\ \gamma_5,\ \gamma^\mu,\ \gamma^\mu\gamma_5,\ \sigma^{\mu\nu}$$

are collectively denoted as Γ^a.

19.8.6 More on Noether currents

Consider the continuity equation for J^μ:

$$\partial_\mu J^\mu = \partial_\mu\bar{\psi}\gamma^\mu\psi + \bar{\psi}\gamma^\mu\partial_\mu\psi$$

substituting from the two Dirac equations yields that the continuity equation is always identically satisfied.

Consider now the continuity equation for J_5^μ :

$$\partial_\mu J_5^\mu = \partial_\mu\bar{\psi}\gamma^\mu\gamma^5\psi + \bar{\psi}\gamma^\mu\gamma_5\partial_\mu\psi$$

now to substitute from the equations of motion a minus sign pops out of the anticommutation of γ^μ, γ_5 and the continuity equation holds **if and only if** the particle is massless.

19.9 Chirality and Chiral projectors

The chiral projectors are the operators

$$P_L = \frac{1}{2}(\mathbb{I} - \gamma_5),\ P_R = \frac{1}{2}(\mathbb{I} + \gamma_5)$$

and their effect is

$$P_L\psi = -\begin{pmatrix}\psi_L\\0\end{pmatrix},\ P_R\psi = \begin{pmatrix}0\\\psi_R\end{pmatrix}.$$

Def. The chirality of a spinor is its eigenvalue under γ_5.

19.10 Massless Dirac equation

Since the chiral phase transformation is a symmetry if and only if $m = 0$, the massless Dirac equation must be considered.

Consider first the Dirac equation in matrix form

$$\begin{pmatrix}-m\mathbb{I} & i(\partial_0 + \sigma^i\partial_i)\\ i(\partial_0 - \sigma^i\partial_i) & -m\mathbb{I}\end{pmatrix}\begin{pmatrix}\psi_L\\\psi_R\end{pmatrix} = 0$$

and this can be expressed as 2 coupled equations

$$\begin{cases} i\left(\partial_0 + \sigma^i\partial_i\right)\psi_R = m\psi_L \\ i\left(\partial_0 - \sigma^i\partial_i\right)\psi_L = m\psi_R \end{cases}:$$

the coupling constant is the mass, so the Dirac equation is reduced to two uncoupled equations for the left and right spinors *if and only if* the particle is massless.

This means that massive fermions oscillate between left and right states, while massless fermions remain in their *chirality*.

Recalling that, for massless particles, $E = |\mathbf{p}| = \partial_0$ and $p_i = \partial_i$, the equations are in the form

$$\frac{\sigma \cdot \mathbf{p}}{|\mathbf{p}|}\psi_R = \psi_R$$

$$\frac{\sigma \cdot \mathbf{p}}{|\mathbf{p}|}\psi_L = -\psi_L :$$

the left and right spinors are eigenstates of the helicity operator with eigenvalues ± 1.

Lesson 14, 20/10

So massless particles do not have $2s + 1$ degrees of freedom, but only the 2 given by helicity, which in this massless case coincides with chirality.

The value of helicity is *not a Lorentz invariant*, since it depends on the momentum $\mathbf{p}$.

19.11 The spin operator

Combining, in the Weyl basis

$$\gamma_5 \sigma^{\mu\nu} = \frac{i}{2}\varepsilon^{\mu\nu\rho\sigma}\sigma_{\sigma\rho}$$

and, recalling the Pauli-Lubanski vector, it is true that

$$W^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}P_\nu M_{\rho\sigma} = -\frac{i}{2}\gamma_5\sigma^{\mu\nu}P_\nu.$$

In the rest frame of the particle, the 4-vectors have value

$$P^\nu = (m, \mathbf{0}),\ W^\mu = (0, -mS^i)$$

and the spin operator is defined to be

$$\Sigma^i = \frac{-2}{m}W^i = \begin{pmatrix} \sigma^i & \mathbb{O} \\ \mathbb{O} & \sigma^i \end{pmatrix}$$

With this definition, helicity is

$$h = \frac{\Sigma_i P^i}{|\mathbf{P}|} = \sigma_i \hat{P}^i$$

and, **for massless particles only**, chirality is

$$\gamma_5 = \frac{\Sigma_i P^i}{|\mathbf{P}|}$$

19.12 Solution to the Dirac Equation

Before solving the equation, it is necessary to introduce the **Dirac representation** of the spinor, in which γ^0 and γ_5 have different forms:

$$\gamma^0 = \begin{pmatrix} \mathbb{I} & \mathbb{O} \\ \mathbb{O} & -\mathbb{I} \end{pmatrix},\ \gamma_5 = \begin{pmatrix} \mathbb{O} & \mathbb{I} \\ \mathbb{I} & \mathbb{O} \end{pmatrix}.$$

And the spinor is interpreted as having the first two components representing *positive energy solutions, particles* and the last two representing *negative energy solutions, antiparticles.*

Consider now the Dirac equation

$$i\gamma^\mu \partial_\mu \psi(x^\mu) = m\psi(x^\mu)$$

this is clearly a plane wave equation, with solutions in the form

$$\begin{cases} \psi_1(x^\mu) = u_s(P^i)e^{-iP_\mu x^\mu} \\ \psi_2(x^\mu) = v_s(P^i)e^{iP_\mu x^\mu} \end{cases}$$

where u_s, v_s are 4-component spinors that depend only on the 3−momentum, since the 0-component is fixed by the dispersion relation

$$E^2 = P_i P^i + m^2$$

which will become important later.

The ψ_1 solution is an inbound plane wave with positive energy, representing a **particle**, while the other is an outbound plane wave with negative energy, representing an **antiparticle**.

The general solution is

$$\psi(x^\mu) = Au_s(P^i)e^{-iP_\mu x^\mu} + Bv_s(P^i)e^{iP_\mu x^\mu}$$

substituting ψ_1 and ψ_2 respectively into the equation yields

$$(i\gamma^\mu P_\mu - m)u_s(P^i) = 0 \to (i\not{P} - m)u_s(P^i) = 0$$

$$(i\gamma^\mu P_\mu + m)v_s(P^i) = 0 \to (i\not{P} + m)v_s(P^i) = 0$$

Rewriting these equations in the rest frame of the particle, $\not{P} = \gamma^0 P_0 = \gamma^0 m$

$$m(\gamma^0 - \mathbb{I})u_s = 0 \iff m\begin{pmatrix} \mathbb{O} & \mathbb{O} \\ \mathbb{O} & -2\mathbb{I} \end{pmatrix}\begin{pmatrix} u \\ \tilde{u} \end{pmatrix} = 0$$

$$m(\gamma^0 + \mathbb{I})v_s = 0 \iff m\begin{pmatrix} \mathbb{O} & \mathbb{O} \\ \mathbb{O} & 2\mathbb{I} \end{pmatrix}\begin{pmatrix} v \\ \tilde{v} \end{pmatrix} = 0$$

The first equation yields the fact that $\tilde{u}(0)$ is the $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ vector, and that $u(0)$ is free: so it is chosen to be either $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ or $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$; the other equation tells a similar tale, so the u, v are

$$u_1(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix},\ u_2(0) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},\ v_1(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix},\ v_2(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Defining

$$\chi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},\ \chi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

they can be expressed as

$$u_s(0) = \begin{pmatrix} \chi_s \\ \mathbf{0} \end{pmatrix},\ v_s(0) = \begin{pmatrix} \mathbf{0} \\ \varphi_s \end{pmatrix}.$$

The normalization is $\bar{u}u = 2m$.

To find the general solution from the rest frame solution, one can perform a boost to a generic reference frame and get the general solution[7], but it can also be found by using the properties of the Clifford Algebra.

Recalling the properties of the Clifford algebra,

$$(\not{P}-m)(\not{P}+m)=(P_\nu P_\mu\gamma^\mu\gamma^\nu-m^2)=(P_\mu P_\nu\frac{1}{2}\underbrace{\{\gamma^\mu,\gamma^\nu\}}_{=2g^{\mu\nu}}-m^2)=(P_\mu P^\mu-m^2)=0,$$

so this operator applied to one of the spinors yields again 0:

$$(\not{P}-m)\underbrace{(\not{P}+m)u_s(0)}_{:=u_s(P^i)}=0\rightarrow u_s(P^i)=(\not{P}+m)u_s(0)$$
$$(\not{P}+m)\underbrace{(-\not{P}+m)v_s(0)}_{:=v_s(P^i)}=0\rightarrow v_s(P^i)=(-\not{P}+m)v_s(0).$$

Expressing the parentheses explicitely:

$$u_s(P^i)=(\not{P}+m)u_s(0)=\begin{pmatrix}(E+m)\mathbb{I} & \sigma_iP^i\\ -\sigma_iP^i & -(E-m)\mathbb{I}\end{pmatrix}\begin{pmatrix}\chi_s\\0\end{pmatrix}=(E+m)\begin{pmatrix}\chi_s\\ \frac{-\sigma_iP^i}{E+m}\chi_s\end{pmatrix}$$
$$v_s(P^i)=(E+m)\begin{pmatrix}\frac{-\sigma_iP^i}{(E+m)}\\ \varphi_s\end{pmatrix}.$$

The orthonormalization relations are

$$\bar{u}_s(P^i)u_r(P^i)=2m\delta_{sr},$$
$$u_s^\dagger(P^i)u_r(P^i)=2E\delta_{sr},$$
$$\bar{v}_s(P^i)v_r(P^i)=-2m\delta_{sr},$$
$$v_s^\dagger(P^i)v_r(P^i)=2E\delta_{sr}.$$

and the spin sums are

$$\sum_{s=1}^{2}u_s\bar{u}_s=\not{P}+m,\ \sum_{s=1}^{2}v_s\bar{v}_s=\not{P}-m.$$

The energy projectors (useful for something) are

$$\left[\Lambda_+(P^i)\right]_{\alpha\beta}=\sum_{2=1}^{2}\frac{u_\alpha^s\bar{u}_\beta^s}{2m}$$

and act like [... anche lambda- vedi foto]

[7] see prof. Boglione handwritten notes for the explicit calculations.

Lesson 15, 21/10

Quantum Field Theory

Up to this point the constructed field theory was purely classical: now the goal is to quantize it.

There are several ways to do so, but the simplest to approach is **Canonical Quantization**, while the better and more difficult approach, **Path Integral Quantization** will be employed for interacting theories in the Foundations of QFT course.

20 Canonical Quantization: general idea

The core idea is to promote the classical fields to *field operators*

$$\phi(x^\mu) \longrightarrow \hat{\phi}(x^\mu),$$

specify its commutation relations with other fields and then express the field as a *Fourier expansion*. The final step is to compute the Hamiltonian of the system and find its eigenstates and eigenvalues.

Schematically, the process can be divided in

- Write the Lagrangian of the system and compute the Hamiltonian of the system;
- Promote the field(s) to operator(s) and set the commutation relations among the fields;
- Expand the field(s) as Fourier Transforms;
- Solve for the eigenvalues and eigenstates of the Hamiltonian.

20.1 Canonical Quantization of the Scalar Field

The Lagrangian of the scalar field is

$$\mathcal{L}(\phi(x^\mu), \partial_\mu\phi(x^\mu)) = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 :$$

consider now the *canonical conjugated field*, defined with a parallelism to conjugated momenta in classic Hamiltonian mechanics

$$\pi(x^\mu) := \frac{\partial\mathcal{L}}{\partial\partial_0\phi} = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi}.$$

Again in analogy with Hamiltonian Mechanics, the Hamiltonian of the system is

$$\mathcal{H} := \pi\dot{\phi} - \mathcal{L} = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}\partial_i\phi\partial^i\phi + \frac{1}{2}m^2\phi^2$$

Note 1 The Hamiltonian formulation *loses* the manifest Lorentz covariance.
Note 2 This Hamiltonian is definite positive, as is expected from a non-interacting theory.

Now ϕ and π are promoted to operators, and their commutator is **defined at fixed time and different space**:

$$\left[\hat{\phi}(t,\mathbf{x}), \hat{\pi}(t,\mathbf{y})\right] = i\hbar\delta^3(\mathbf{x}-\mathbf{y})$$

again in analogy to the quantization of Hamiltonian mechanics for point-like particles. There is an explanation for the result of this commutator, connected with Noether charges that will not be covered in the lectures, but is present in the notes of prof. Boglione.

The other commutation relations are trivial

$$\left[\hat{\phi}(t,\mathbf{x}),\hat{\phi}(t,\mathbf{y})\right]=0,\ [\hat{\pi}(t,\mathbf{x}),\hat{\pi}(t,\mathbf{y})]=0$$

but have a deep implication: if a measurement is taken at the same time, but in different places, their result will be **independent**, salvaging causality in the theory.

Consider now the Fourier Transform of $\hat{\phi}$:

$$\hat{\phi}(x^\mu)=\frac{1}{(2\pi)^{\frac{3}{2}}}\int \mathrm{d}^4k\delta(k^2-m^2)\vartheta(k^0)\left[\hat{A}(\mathbf{k})e^{-ikx}+\hat{A}^\dagger(\mathbf{k})e^{ikx}\right]$$

where $k^2=k_\mu k^\mu$ and $kx=k_\mu x^\mu$
Notes

- The measure of integration is Lorentz-invariant;
- k^μ plays the role of the 4-momentum, so $k^0=E_k$;
- The Heaviside ϑ guarantees $k^0=E_k\geq 0$;
- The Dirac δ guarantees the mass-shell dispersion relations and fixes one of the components of k^μ: for convention, k^0 is used.

Using the δ, an integration over k^0 is performed, recalling

$$\delta(f(k))=\sum_a\frac{\delta(k-a)}{|f'(a)|}$$

where a are the simple zeroes of $f(k)$, the argument of the δ in the F.T. has poles at

$$k^2=m^2\iff k^0=\pm\sqrt{\mathbf{k}^2+m^2}$$

and so

$$\delta(k^2-m^2)=\frac{\delta(k^0-\sqrt{\mathbf{k}^2+m^2})}{2k_0}+\frac{\delta(k^0+\sqrt{\mathbf{k}^2+m^2})}{2|k^0|}$$

the second contribution is not admissible, since the ϑ filters out the $k^0<0$ values:

$$\delta(k^2-m^2)\vartheta(k^0)=\frac{\delta(k^0-\omega_k)}{2\omega_k},\ \text{with}\ \omega_k=+\sqrt{\mathbf{k}^2+m^2}$$

The F.T. becomes

$$\hat{\phi}(x^\mu)=\int\frac{\mathrm{d}^3k}{(2\pi)^{\frac{3}{2}}}\frac{1}{2\omega_k}\left[\hat{A}(\mathbf{k})e^{-ikx}+\hat{A}^\dagger(\mathbf{k})e^{ikx}\right]$$

and, defining

$$\hat{a}^{(+)}=\frac{\hat{A}^{(+)}}{\sqrt{2\omega_k}}$$

one gets

$$\boxed{\hat{\phi}(x^\mu)=\int\frac{\mathrm{d}^3k}{(2\pi)^{\frac{3}{2}}}\frac{1}{\sqrt{2\omega_k}}\left[\hat{a}(\mathbf{k})e^{-ikx}+\hat{a}^\dagger(\mathbf{k})e^{ikx}\right]}$$

and the conjugated field

$$\boxed{\hat{\pi}(x^\mu)=\dot{\hat{\phi}}=\int\frac{\mathrm{d}^3k}{(2\pi)^{\frac{3}{2}}}\frac{1}{\sqrt{2\omega_k}}i\omega_k\left[-\hat{a}(\mathbf{k})e^{-ikx}+\hat{a}^\dagger(\mathbf{k})e^{ikx}\right].}$$

Since now all the information on the field is encoded in $\hat{a}^{(+)}$, it is necessary to find out what are these operators: consider the commutation relation

$$\left[\hat{\phi}(t,\mathbf{x}),\hat{\pi}(t,\mathbf{y})\right]=\int\frac{\mathrm{d}^3k}{(2\pi)^{3/2}(2\omega_k)^{1/2}}\int\frac{i\omega_{k'}\mathrm{d}^3k'}{(2\pi)^{3/2}(2\omega_{k'})^{1/2}}\left[(\hat{a}_ke^{-ikx}+a_k^\dagger e^{ikx}),(-a_{k'}e^{-ik'y}+a_{k'}^\dagger e^{ik'y})\right]$$

it is NOT easy to show that for this commutation relation to yield $i\delta^3(\mathbf{x}-\mathbf{y})$, the a operators must obey

$$\boxed{[\hat{a}_k,\hat{a}_{k'}]=0,\ \left[\hat{a}_k^\dagger,\hat{a}_{k'}^\dagger\right]=0,\ \left[\hat{a}_k,\hat{a}_{k'}^\dagger\right]=\delta^3(\mathbf{k}-\mathbf{k}').}$$

these operators obey the **same algebra of the creation and annihilation operators for the quantum harmonic oscillator** and they are interpreted as operators that **create and destroy local ripples in the fields**, which are **the particles we see.**

Now, the Hamiltonian is computed with the F.Ts of the fields

$$H=\int\mathrm{d}^3x\mathcal{H}=\frac{1}{2}\int\mathrm{d}^3x\left(\hat{\pi}^2+\partial_i\hat{\phi}\partial^i\hat{\phi}+m^2\hat{\phi}^2\right)=$$
$$=\frac{1}{2}\int\mathrm{d}^3x\int\frac{\mathrm{d}^3k\mathrm{d}^3k'}{(2\pi)^3\sqrt{2\omega_k}\sqrt{2\omega_{k'}}}\Big[-(\omega_k\omega_{k'}+\mathbf{k}\cdot\mathbf{k}')(\hat{a}_ke^{-ikx}-\hat{a}_k^\dagger e^{ikx})(\hat{a}_{k'}e^{-ik'x}-\hat{a}_{k'}^\dagger e^{ik'x})+$$
$$+m^2(\hat{a}_ke^{-ikx}+\hat{a}_k^\dagger e^{ikx})(\hat{a}_{k'}e^{-ik'x}+\hat{a}_{k'}^\dagger e^{ikx})\Big].$$

Integrating over x only touches the 3-vector part of the exponentials, yielding Dirac $\delta^3(\mathbf{k}\pm\mathbf{k}')$ and greatly simplifying matters.

$$H=[...]=\frac{1}{2}\int\mathrm{d}^3k\omega_k(\hat{a}_k\hat{a}_k^\dagger+\hat{a}_k^\dagger\hat{a}_k)$$

The commutation relation with $\mathbf{k}=\mathbf{k}'$ yields a problematic result

$$\left[\hat{a}(k),\hat{a}^\dagger(k)\right]=\delta^3(0)\to\infty$$

Using the commutation relation in H yields

$$H=\int\mathrm{d}^3k\omega_k\left(\hat{a}_k\hat{a}_k^\dagger+\frac{1}{2}\delta^3(0)\right):$$

the form of this Hamiltonian density is in the form of that of the familiar quantum harmonic oscillator, with the key difference that the 0-point energy is $\delta^3(0)$; this enforces the idea that the scalar field can be though of as an infinite sum of Harmonic Oscillators, so its ground state energy is the divergent series

$$\sum_{i=0}^{\infty}\frac{1}{2}\omega.$$

This problem with the δ can be resolved by setting the **0-point energy to 0**, since this is always allowed in physics; to do so, the Hamiltonian must be redefined so that the annihilation operators are on the right: this is called **normal ordering**

$$N(H)=:H:=\int\mathrm{d}^3k\omega_k\hat{a}_k^\dagger\hat{a}_k$$

and it consists in putting the annihilation operators, $\hat{a}_k$ in this case, on the right side of the Hamiltonian density $\mathcal{H}$.

Lesson 16, 22/10

The momentum operator can be derived in a similar fashion to the Hamiltonian:

$$\hat{P}^i = \int \mathrm{d}^3x \hat{T}^i{}_0 = \int \mathrm{d}^3x \hat{\pi}\partial^i\hat{\phi}$$

substituting in the F.Ts of the two fields and using the commutation relations; using the normal ordering this yields

$$:\hat{P}^i := \int \mathrm{d}^3k k^i \hat{a}_k^\dagger \hat{a}_k$$

which is similar in form to $\hat{H}$, since $\omega_k = k^0$.

20.2 Fock Space

The space of the states on which the operators are defined is a "reordered" Hilbert Space:

$$\mathcal{H}_{\text{Fock}} = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus ...$$

where $\mathcal{H}_1$ is the Hilbert subspace in which the basis is made by all the stated with 1 particle and

$$\mathcal{H}_i = \underbrace{\mathcal{H}_1 \otimes \mathcal{H}_1 \otimes ...}_{i \text{ times}}.$$

The $\mathcal{H}_0$ space is trivial, as it contains only the vacuum state $|0\rangle$.

The creation and annihilation operators add or remove particles from the state:

$$\hat{a}(\mathbf{k})|0\rangle = 0$$
$$\hat{a}^\dagger(\mathbf{k})|0\rangle := |\mathbf{k}\rangle \in \mathcal{H}_1$$
$$|\mathbf{k_1}, \mathbf{k_2}, ...\mathbf{k_n}\rangle = \hat{a}^\dagger(\mathbf{k_1}); \hat{a}^\dagger(\mathbf{k_2})...\hat{a}^\dagger(\mathbf{k_n})|0\rangle,$$

which represents a state with n particles of momentum $\mathbf{k_i}$: if these momenta are equal, the shorthand notation is

$$|n(\mathbf{k})\rangle = \frac{(\hat{a}^\dagger(\mathbf{k}))}{\sqrt{n!}}|0\rangle.$$

The number operator is defined as

$$\hat{n}(\mathbf{k}) = \hat{a}^\dagger(\mathbf{k})\hat{a}(\mathbf{k})$$

and its eigenstates are $|n(\mathbf{k})\rangle$:

$$\hat{n}(\mathbf{k})|n(\mathbf{k})\rangle = n|n(\mathbf{k})\rangle.$$

The orthogonality condition

$$\langle \mathbf{k}|\mathbf{k}'\rangle = \delta^3(\mathbf{k} - \mathbf{k}')$$

is not Lorentz invariant, so the normalization is set to be

$$\int \mathrm{d}^3k \frac{1}{(2\pi)^3 2\omega_k}|k\rangle\langle k| = \mathbb{I}$$

where $|k\rangle$ is the state with 4-momentum k and differs from $|\mathbf{k}\rangle$ only by a normalization

$$|k\rangle = (2\pi)^{\frac{3}{2}}\sqrt{2\omega_k}|\mathbf{k}\rangle.$$

How does the field operator act on the vacuum?

$$\hat{\phi}|0\rangle = \int \mathrm{d}^3k\mathcal{N}\left[\hat{a}_k|0\rangle e^{-ikx} + \hat{a}_k^\dagger|0\rangle e^{ikx}\right] = \int \mathrm{d}^3k\mathcal{N}e^{ikx}|\mathbf{k}\rangle$$

this can be interpreted as the field exciting the vacuum creating a particle for each and every oscillating mode (given by $\mathbf{k}$)

Attaching a $\langle k'|$ on the left yields a transition amplitude between $\mathbf{k}$ and $\mathbf{k}'$

$$\left\langle k' \right| \hat{\phi}(x^\mu) \left|0\right\rangle = \int \mathrm{d}^3k \frac{\sqrt{2\omega_{k'}}}{\sqrt{2\omega_k}} e^{ikx} \delta^3(\mathbf{k} - \mathbf{k}') = e^{ik'x}$$

Note With this formalization, the Bose-Einstein Statistic for scalar particles comes for free: consider the creation of two particles with different momenta

$$a^\dagger(k')a^\dagger(k) \left|0\right\rangle = a^\dagger(k)a^\dagger(k') \left|0\right\rangle \neq 0,$$

since $\left[a^\dagger(k), a^\dagger(k')\right] = 0$. The exchange in the order of creation of the particles is irrelevant (this will not be the case for the Dirac Field). Moreover the commutation relation does not forbid the successive creation of n particles with the same momentum k^i:

$$\forall n,\ (a^\dagger(k))^n \left|0\right\rangle \neq 0$$

effectively resembling the Bose-Einstein statistic.

20.3 Canonical Quantization of the charged scalar field

The Lagrangian of the charged scalar field is

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi \phi^\dagger;$$

the conjugated fields are two:

$$\pi_\phi = \frac{\partial \mathcal{L}}{\partial \partial_0 \phi} = \dot{\phi}^\dagger$$
$$\pi_{\phi^\dagger} = \frac{\partial \mathcal{L}}{\partial \partial_0 \phi^\dagger} = \dot{\phi}$$

The Hamiltonian density is defined as

$$\mathcal{H} = \pi_\phi \dot{\phi} + \pi_{\phi^\dagger} \dot{\phi}^\dagger - \mathcal{L} = \dot{\phi}^\dagger \dot{\phi} - \partial_i \phi^\dagger \partial^i \phi + m^2 \phi^\dagger \phi :$$

now the fields are promoted to operators, with commutation relations

$$[\phi(t,x), \pi_\phi(t,y)] = \left[\phi^\dagger(t,x), \pi_{\phi^\dagger}\right] = i\delta^3(\mathbf{x} - \mathbf{y})$$

and all the other are zero.

The Fourier transforms of the two fields are

$$\phi(x^\mu) = \int \frac{\mathrm{d}^3k}{(2\pi)^{\frac{3}{2}}\sqrt{2\omega_k}} \left[a_k e^{-ikx} + b_k^\dagger e^{ikx}\right]$$
$$\phi^\dagger(x^\mu) = \int \frac{\mathrm{d}^3k}{(2\pi)^{\frac{3}{2}}\sqrt{2\omega_k}} \left[a_k^\dagger e^{ikx} + b_k e^{-ikx}\right]$$

Note There are two sets of operators because the charged scalar field is composed of two different real scalar fields.

And so, a are associated to particles, while b to antiparticles.

The canonical commutation relations yield

$$\left[a_k, a^\dagger_{k'}\right] = \left[b_k, b^\dagger_{k'}\right] = \delta^3(\mathbf{k}-\mathbf{k}')$$

and all others are zero: this is the algebraic structure of two decoupled harmonic oscillators.

With a big ugly calculation, the normal ordered Hamiltonian turns out to be

$$:H: = \int \mathrm{d}^3k\omega_k \left(a^\dagger_k a_k + b^\dagger_k b_k\right) = \int \mathrm{d}^3x\omega_k \left(n^a(\mathbf{k}) + n^b(\mathbf{k})\right).$$

20.3.1 Noether currents and charges

Recall that the charged scalar field Lagrangian is invariant under the transformation

$$\phi \to \phi' = e^{i\alpha}\phi :$$

the Noether current associated to this symmetry is an operator and it is

$$J^\mu = \phi^\dagger \partial_\mu \phi - \partial_\mu \phi^\dagger \phi$$

and the conserved charge is

$$Q = \int \mathrm{d}^3x J^0 = \int \mathrm{d}^3x (\phi^\dagger \dot{\phi} - \dot{\phi}^\dagger \phi)$$

replacing the fields' F.Ts in this expression yields

$$Q = \int \mathrm{d}^3k \left(a^\dagger_k a_k - b^\dagger_k b_k\right) = \int \mathrm{d}^3k \left(n^a(\mathbf{k}) - n^b(\mathbf{k})\right).$$

This can be interpreted as the *total charge* of the system, with the "a" particles have positive charge, while the "b" particles have negative charge: the conserved quantity is then the total (Abelian) charge and this conservation means that creating an "a" particle implies creating a "b" particle.

20.4 Canonical Quantization of the Dirac Field

The Lagrangian for the spinorial field is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$$

The conjugated fields are

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\psi^\dagger, \qquad \pi_{\bar{\psi}} = \frac{\partial \mathcal{L}}{\partial \dot{\bar{\psi}}} = 0$$

and so the Hamiltonian density is

$$\mathcal{H} = \pi_\psi \dot{\psi} + \pi_{\bar{\psi}} \dot{\bar{\psi}} - \mathcal{L} = \bar{\psi}(-i\gamma^i \partial_i + m)\psi$$

using the Dirac equation to express $i\gamma^i\partial_i = m - i\gamma^0\partial_0$ yields

$$\mathcal{H} = i\psi^\dagger \dot{\psi}.$$

Now the commutation relations need to be set between $\psi, \bar{\psi}$ and their conjugated fields: since these are fermionic fields, anticommutators are expected (since a theory with commutators does not conserve energy):

$$\{\psi_s(t,x), \pi_{\psi,r}(t,y)\} = i\delta^3(\mathbf{x}-\mathbf{y})\delta_{rs} \to \left\{\psi_s(t,x), \psi^\dagger_r(t,y)\right\} = \delta^3(\mathbf{x}-\mathbf{y})\delta_{rs}, \text{ with } r,s = 1,2.$$

Note Since the fields are spinors, these anticommutation relations also depend on the spin of the field.

Lesson 17, 26/10

Note Since from Quantum Mechanics spin-$\frac{1}{2}$ particles obey the Fermi-Dirac statistic, we expect anti-symmetry for the Dirac field and this is provided by *anti*commutators.

As before, the fields are expanded in their F.Ts:

$$\psi(x^\mu) = \int \frac{\mathrm{d}^3 p}{(2\pi)^{\frac{3}{2}}\sqrt{2E_p}} \sum_{s=1}^{2} \left[u_s(\mathbf{p}) a_s(\mathbf{p}) e^{-ipx} + v_s(\mathbf{p}) b_s^\dagger(\mathbf{p}) e^{ipx} \right]$$

where $px = P_\mu x^\mu$, $P^0 = E_p$ and u_s, v_s are the spinors defined earlier in the solution of the Dirac equation.

It can be shown that the field anticommutation relation in terms of the operators a, b is

$$\left\{ \psi_s(t,x), \psi_r^\dagger(t,y) \right\} = \delta^3(\mathbf{x}-\mathbf{y})\delta_{rs} \iff \left\{ a_s(\mathbf{p}), a_r^\dagger(\mathbf{p}') \right\} = \left\{ b_s(\mathbf{p}), b_r^\dagger(\mathbf{p}') \right\} = \delta^3(\mathbf{p}-\mathbf{p}')\delta_{rs}$$

and the other are zero.

From this expansion of the field, $\psi^\dagger$ and $\dot{\psi}$ can be built and the Hamiltonian turns out to be

$$H = \int \mathrm{d}^3 x \mathcal{H} = i \int \mathrm{d}^3 x \psi^\dagger \dot{\psi}$$

the strategy to integrate is to integrate over $\mathbf{x}$, yielding $\delta^3(\mathbf{p}-\mathbf{p}')$ and then using this to apply the orthogonality relations of the spinors $u, v, u^\dagger, v^\dagger$:

$$H = \int \mathrm{d}^3 p \sum_{s,r=1}^{2} \frac{1}{2} \left[u_s^\dagger(\mathbf{p}) u_r(\mathbf{p}) a_s^\dagger(\mathbf{p}) a_r(\mathbf{p}) - v_s^\dagger(\mathbf{p}) v_s(\mathbf{p}) v_r(\mathbf{p}) b_s(\mathbf{p}) b_r^\dagger(\mathbf{p}) \right]$$

since

$$u_s^\dagger u_r = v_s^\dagger v_r = 2E_p \delta_{rs},$$

and

$$b_s b_s^\dagger = -b_s^\dagger b_s + \delta^3(0),$$

after taming the infinite offset, the Hamiltonian finally is

$$N(H) = \int \mathrm{d}^3 p E_p \sum_s \left[a_s^\dagger(\mathbf{p}) a_s(\mathbf{p}) + b_s^\dagger(\mathbf{p}) b_s(\mathbf{p}) \right].$$

Note Had the relations between ψ, π_ψ been commutators, the sign of the $b_s^\dagger b_s$ would have changed, giving way to a theory without *definite positive* energy, which is non-physical.

The operators have a similar physical interpretation: $a_s^\dagger(\mathbf{p})$ creates a fermion with spin s and momentum $\mathbf{p}$, while $b_s^\dagger(\mathbf{p})$ creates an antifermion with same spin and momentum.

Note The form of the Hamiltonian is remarkable: the terms $a_s^\dagger a_s$, $b_s^\dagger b_s$ are number operators, so the total energy H is the energy E_p multiplied by the total number of particles and antiparticles of momentum p^i and integrated over all the possible values of p^i.

Note This formalization has a **built-in Pauli exclusion principle,** since applying two creation operators with same quantum number s and same momentum p to the vacuum yields

$$a_s^\dagger(p) a_s^\dagger(p) \left|0\right\rangle = -a_s^\dagger(p) a_s^\dagger(p) \left|0\right\rangle \equiv 0$$

while, if $r \neq s$ or $p \neq p'$,

$$a_s^\dagger(p) a_r^\dagger(p') \left|0\right\rangle = -a_r^\dagger(p') a_s^\dagger(p) \left|0\right\rangle \neq 0$$

since $\left\{a_s^\dagger(\mathbf{p}), a_r^\dagger(\mathbf{p}')\right\} = 0$: this closely resembles the Fermi-Dirac Statistic and is a direct consequence of the choice of anticommutators over commutators.

Using again normal ordering and the F.Ts of the fields, the momentum operator is

$$P^i = \int \mathrm{d}^3x \psi^\dagger(-i\partial^i)\psi \to N(P^i) = \int \mathrm{d}^3p p^i \sum_s (a_s^\dagger a + b_s^\dagger b_s).$$

Again this is the number of particles and antiparticles of momentum p^i, multiplied by their momentum and integrated over all possible values of it.

20.4.1 Noether current and charges

It was shown that the phase transformation

$$\psi \to \psi' = e^{i\alpha}\psi$$

is a symmetry for the Lagrangian and its associated Noether current is

$$J^\mu = \bar{\psi}\gamma^\mu\psi$$

with charge

$$Q = \int \mathrm{d}^3x J_0 = \int \mathrm{d}^3x \psi^\dagger\psi$$

replacing the F.Ts into the integral (with the same *caveat* of integrating first over x) yields

$$Q = \int \mathrm{d}^3p \sum_s a_s^\dagger a_s + b_s b_s^\dagger$$

the normal ordering yields

$$N(Q) = \int \mathrm{d}^3p \sum_s \left[a_s^\dagger a_s - b_s^\dagger b_s\right]$$

this implies that the number of fermions - the number of antifermions **is conserved**: so every time a fermion is created, an antifermion must also be created [ragionamento su H e P, perso...].

Using Noether's theorem on Lorentz invariance of the Dirac field, the Energy-momentum tensor turns out to be

$$T^{\mu\nu} = i\bar{\psi}\gamma^\mu\partial^\nu\psi$$

and the angular momentum becomes

$$M^{\lambda\mu\nu} = ... = i\bar{\psi}\gamma^\lambda(\underbrace{x^\mu\partial^\nu - x^\nu\partial^\mu}_{\text{orbital ang mom}} \underbrace{-\frac{i}{2}\sigma^{\mu\nu}}_{\text{spin}})$$

[RIVEDI!]

21 The Propagator

Until now, time was fixed, so the next natural step is to consider the (space-)time evolution of the fields.

The propagator in QM is defined as

$$\psi(\mathbf{x}, t) = \int \mathrm{d}^3y K(\mathbf{y}, t', \mathbf{x}, t)\psi(\mathbf{y}, t')$$

Def. The free propagator of a field ϕ is the function

$$G(x^\mu, y^\nu) = \vartheta(x^0 - y^0) \langle 0| \phi(x^\mu)\phi^\dagger(y^\nu) |0\rangle + \vartheta(y^0 - x^0) \langle 0| \phi(y^\mu)\phi^\dagger(x^\mu) |0\rangle$$

The first sandwich part can be interpreted as follows: $\phi^\dagger$ creates a probe particle at spacetime y^ν, the system is then left to evolve, until at the spacetime point x^μ the particle is destroyed by ϕ; the Heaviside theta ensures the particle is destroyed only after it is created. The second addend is the exact same, but with the creation and destruction of an antiparticle.

The propagator is the **transition probability amplitude** of returning to the ground state, which, in the case of free theories, is the vacuum $|0\rangle$.

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Note The function is named G because it is a **Green function** of the operators in the equations of motion of the fields, as will be proven.

21.1 Naming conventions

The Feynman propagator is the function Δ_F, which is defined so that

$$i\Delta_f = G(x^\mu, y^\nu);$$

another notation is to indicate the two parts of G as, respectively, $D(x-y)$, $D(y-x)$:

$$G(x,y) = \vartheta(x^0 - y^0)D(x-y) + \vartheta(y^0 - x^0)D(y-x).$$

Another important notation is that of the *time ordering operator* T:

$$T\psi(x)\chi(y) = \begin{cases} \psi(x)\chi(y) & \text{if } x^0 > y^0 \\ \pm\chi(y)\psi(y) & \text{if } y^0 > x^0 \end{cases}$$

where the $+$ is used for scalar fields, while the $-$ for Dirac fields.

This also has a graphical interpretation

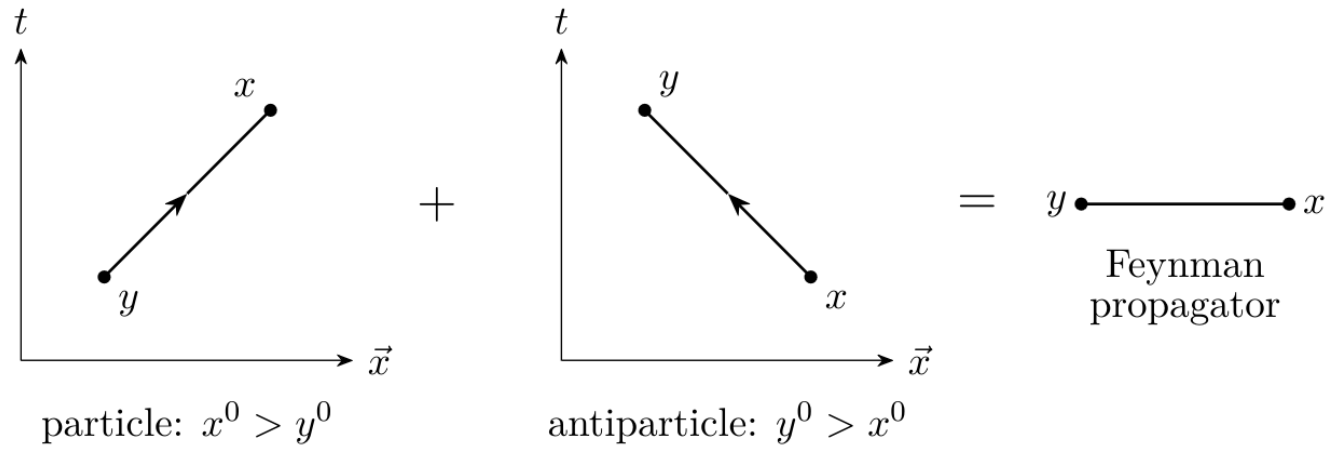


Fig. 21.1: From Prof. Boglione's notes

21.2 Propagator for the scalar field

The goal is to find the explicit form of the propagator for the scalar field

$$\phi(x^\mu) = \int \frac{\mathrm{d}^3p}{(2\pi)^{3/2}\sqrt{2E_p}} \left[a_p e^{-ipx} + a_p^\dagger e^{ipx}\right].$$

Since the field is real, the propagator becomes

$$i\Delta_F = \vartheta(x^0 - y^0) \langle 0| \phi(x)\phi(y) |0\rangle + \vartheta(y^0 - x^0) \langle 0| \phi(y)\phi(x) |0\rangle$$

consider

$$\phi(y) |0\rangle = \int \frac{\mathrm{d}^3p'}{(2\pi)^{\frac{3}{2}}\sqrt{2E_{p'}}} e^{ip'y} \left|\mathbf{p}'\right\rangle$$

where $a_{p'} |0\rangle = 0$ and $a_{p'}^\dagger |0\rangle = |\mathbf{p}'\rangle$. Similarly

$$\langle 0| \phi(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^{\frac{3}{2}}\sqrt{2E_p}} e^{-ipx} \langle \mathbf{p}|$$

and so

$$D(x-y) = \langle 0| \phi(x)\phi(y) |0\rangle = \int \frac{\mathrm{d}^3p\mathrm{d}^3p'}{(2\pi)^3 2\sqrt{E_pE_{p'}}} e^{-ipx+ip'y} \left\langle \mathbf{p}|\mathbf{p}'\right\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p} e^{-ip(x-y)}.$$

Similarly, the second term is

$$D(y-x) = \langle 0| \phi(y)\phi(x) |0\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p} e^{-ip(y-x)}.$$

and so

$$i\Delta_F = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p} \left[\vartheta(x^0 - y^0) e^{-ip(x-y)} + \vartheta(y^0 - x^0) e^{ip(x-y)}\right]$$

Letting $z^\mu = x^\mu - y^\mu$, the needed functions this is

$$i\Delta_F = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p} \left[\vartheta(z^0) e^{-ipz} + \vartheta(y^0 - x^0) e^{ipz}\right]$$

To compute the integral explicitely, the theta function integral representation is needed.

21.2.1 Integral representation of the theta function

The Theta function is defined as

$$\vartheta(z^0) = \begin{cases} 1 & \text{if } z^0 \geq 0 \\ 0 & \text{if } z^0 \leq 0 \end{cases}.$$

Consider a complex function with a simple pole in $i\varepsilon$, with $\varepsilon > 0$:

$$\frac{e^{i\tau z_0}}{\tau - i\varepsilon}$$

then its integral over the real axis

$$I(z^0) = \lim_{\varepsilon \to 0^+} \frac{1}{2\pi i} \int_{-\infty}^{\infty} \mathrm{d}\tau \frac{e^{i\tau z^0}}{\tau - i\varepsilon}$$

can be computed using Jordan's lemma, depending on the sign of z^0: if it is positive the contour is closed in $\mathrm{Im}(\tau) > 0$ and the residue is included in the contour and the integral is nonzero, while if z^0 is negative, the contour must be closed in $\mathrm{Im}(\tau) < 0$ and the integral is zero, so

$$I(z^0) = \vartheta(z^0) = \begin{cases} 1 & \text{if } z^0 \geq 0 \\ 0 & \text{if } z^0 \leq 0 \end{cases}.$$

Inserting this representation and returning to the full Lorentz-invariant measure of integration yields

$$i\Delta_F = \frac{1}{(2\pi)^3} \int d^4p \vartheta(p^0)\delta(p^2 - m^2) \left[\vartheta(z^0)e^{-ipz} + \vartheta(-z^0)e^{ipz}\right] =$$
$$= \lim_{\varepsilon \to 0^+} \frac{1}{i(2\pi)^4} \int \mathrm{d}^4p \int_{-\infty}^{\infty} \mathrm{d}\tau \vartheta(p^0)\delta(p^2 - m^2) \left\{ \frac{e^{-i(pz - \tau z^0)}}{\tau - i\varepsilon} + \frac{e^{i(pz - \tau z^0)}}{\tau - i\varepsilon} \right\}$$

The strategy here is to find a new variable so that the second exponential has the same sign as the first. Letting

$$p'^0 = p^0 - \tau,\ \mathbf{p'} = \mathbf{p},\ \omega_{p'}^2 := \mathbf{p'}^2 + m^2 \implies \omega_{p'}^2 - (p'^0)^2 = m^2 - (p')^2$$

modifies the mass-shell:

$$\delta(p^2 - m^2) = \delta((p'^0 + \tau)^2 - \omega_{p'}^2) = \frac{\delta(p'^0 + \tau + \omega_{p'})}{2\omega_{p'}} + \frac{\delta(p'^0 + \tau - \omega_{p'})}{2\omega_{p'}}$$

[something with the product with $\vartheta(p^0)$]

$$i\Delta_F = \frac{1}{i(2\pi)^4} \int \mathrm{d}^4p' \int_{-\infty}^{\infty} \mathrm{d}\tau e^{-ip'z} \left\{ \frac{\delta(p'^0 + \tau + \omega_{p'})}{2\omega_{p'}} + \frac{\delta(p'^0 + \tau - \omega_{p'})}{2\omega_{p'}} \right\} \left\{ \frac{\vartheta(p'^0 + \tau)}{\tau - i\varepsilon} + \frac{\vartheta(p'^0 - \tau)}{\tau - i\varepsilon} \right\}$$

2 out of the 4 products contain incompatible products of δ and ϑ
{vedi foto per il continuo dei conti + note boglione}
The final result is

$$\boxed{\Delta_F(x - y) = \lim_{\varepsilon \to 0^+} \frac{1}{(2\pi)^4} \int \mathrm{d}^4p \frac{e^{-ip(x-y)}}{p^2 - m^2 + i\varepsilon}}$$

Its Fourier transform in the momentum space is manifest

$$\Delta_F(p) = \lim_{\varepsilon \to 0^+} \frac{1}{p^2 - m^2 + i\varepsilon}$$

Note The $i\varepsilon$ at the denominators come from the ϑ, so they guarantee causality.

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21.2.2 Propagator as a Green function

In general, **the propagator of a free theory is a Green function of operator generating the equation of motion of the theory**.

The expression explicitely derived is the propagator of a *scalar field*, so Δ_F should be a Green function for the Klein-Gordon operator, i.e.

$$(\Box + m^2)\Delta_F(x - y) = -\delta^4(x - y)$$

this is true, since the $\Box$ acts only on the exponential inside the integral as

$$\partial_\mu \partial^\mu e^{-ip(x-y)} = -p^2 e^{-ip(x-y)}$$

so

$$(\Box + m^2)\Delta_F = (-p^2 + m^2)\Delta_F = -\delta^4(x-y)^8$$

Note There are more Green functions for eack operator, depending on the way the integral is regularized, namely on the choice of deformation of the contour integral: only one of these choices respects causality and it is the one prescribed by Feynman, passing below the $p^0 = -\omega_p$ and above the $p^0 = \omega_p$ singularities, which is equivalent to the choice of adding $i\varepsilon$ to the denominator, with $\varepsilon > 0$, since this moves the singularities by $\pm i\frac{\varepsilon}{2\omega_p}$ respectively.
This integration over p^0, computed with the residues in the poles according to the sign of $z^0 = x^0 - y^0$, yields

$$i\Delta_F = \vartheta(z^0)\int \frac{\mathrm{d}^3p}{(2\pi)^3 2\omega_p} e^{-ipz} + \vartheta(-z^0)\int \frac{\mathrm{d}^3p}{(2\pi^3 2\omega_p)} e^{ipz}$$

which is the expression found earlier by using the F.Ts of the fields and imposing time ordering.

21.3 The propagator and Causality

From Special Relativity, it is known that if two events are separated by a *space-like interval*

$$(x-y)^2 < 0$$

there exist a Lorentz Transformation that switches the sign of $x^0 - y^0$: the theory should not permit a measurement of the field at a point x to influence a measurement at any point y separated by a space-like interval from x, this means that

$$[\phi(x), \phi(y)] = 0, \text{ if } (x-y)^2 < 0$$

Expanding the commutator yields

$$[\phi(x), \phi(y)] = \int \frac{\mathrm{d}^3k \mathrm{d}^3k'}{(2\pi)^2 2\sqrt{\omega_k \omega_{k'}}} \left[a_k e^{-ikx} + a_k^\dagger e^{ikx}, a_{k'} e^{-ik'y} + a_{k'}^\dagger e^{ik'y}\right]$$

Using the a_k's commutation relations and simplifying the resulting δ^3's yields

$$[\phi(x), \phi(y)] = \int \frac{\mathrm{d}^3k}{(2\pi)^3 2\omega_k} \left(e^{-ik(x-y)} - e^{ik(x-y)}\right) = D(x-y) - D(y-x)$$

where $D(x-y)$ was earlier defined with the propagator.

To show that this is zero for space-like intervals, one might consider a frame of reference in which $x^0 - y^0 = 0$ and $|\mathbf{x} - \mathbf{y}| = r$: this yields

$$\lim_{r\to\infty} D(x-y) \sim e^{-mr}$$

that is not zero, so the key to the solution is that *there exist a continuous Lorentz transformation* that transforms $x - y \to y - x$ and so the two functions are exactly the same, cancelling each other out.

For time-like intervals, i.e.

$$(x-y)^2 > 0,$$

there always exist a transformation so that the 3-vectors are equal, so the limit is

$$\lim_{t\to\infty} D(x-y) \sim e^{-imt}$$

which does not even go to zero; moreover, to invert $x-y$ as a time-like interval a time-reversal transformation is needed, which is not permitted in the theory, making it impossible.

[8] non chiaro, prova a farlo esplicitamente come lo aveva fatto Tateo

21.4 The Dirac Field Propagator

The Dirac propagator is in the same form as the scalar, with the caveats $\phi^\dagger \to \bar{\psi}$ and $\bar{\psi}\psi = -\psi\bar{\psi}$:

$$iS_F(x-y)_{ab} = \vartheta(x^- y^0) \langle 0| \psi_a(x)\bar{\psi}_b(y) |0\rangle - \vartheta(y^0 - x^0) \langle 0| \bar{\psi}_b(y)\psi_a(x) |0\rangle$$

Note ψ is a column spinor, while $\bar{\psi} \propto \psi^\dagger$ is a row spinor, so S_F is **a matrix**, like $\psi\bar{\psi}$.

Like before, replacing the F.Ts of the fields yields only one non-zero addend:

$$\langle 0| \psi(x)\bar{\psi}(y) |0\rangle = \int \frac{\mathrm{d}^3p\mathrm{d}^3p'}{(2\pi)^3 2\sqrt{E_p E_{p'}}} \sum_{r,s} u_r(p')\bar{u}_s(p)e^{-ipx+ipy} \langle 0| a_r(p')a_s^\dagger(p) |0\rangle$$

and the other term turns out to be

$$\langle 0| \bar{\psi}(y)\psi(x) |0\rangle = \int \frac{\mathrm{d}^3p\mathrm{d}^3p'}{(2\pi)^3 2\sqrt{E_p E_{p'}}} \sum_{r,s} v_r(p')\bar{v}_s(p)e^{-ip'y+ipx} \langle 0| b_r(p')b_s^\dagger(p) |0\rangle$$

Note In the integrals the terms $\langle 0| aa^\dagger |0\rangle$ and the similar with b are fermionic and anti-fermionic contributions.
Recalling the anticommutation relations of a, b and the spin sums

$$\sum_s u_s(p)\bar{u}_s(p) = \left(\not{p} + m\right)$$

$$\sum_s v_s(p)\bar{v}_s(p) = \left(\not{p} - m\right)$$

the final result is

$$S_F(x-y) = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p} \left[(\not{p}+m)e^{-ip(x-y)}\vartheta(x^0-y^0) - (\not{p}-m)e^{ip(x-y)}\vartheta(y^0-x^0)\right]$$

Note it can[9] be proven that

$$S_F(x-y) = (i\not{\partial} + m)\Delta_F(x-y) = \lim_{\varepsilon\to 0} \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-ip(x-y)} \frac{\not{p}+m}{p^2 - m^2 + i\varepsilon}.$$

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Making the explicit calculations is not difficult: the derivatives of the ϑ simplify for symmetry reasons, so $\not{\partial}$ acts only on the exponentials

$$i\not{\partial}_x e^{-ipz} = \not{p}e^{-ipz}$$

and the result follows.

Note The Dirac propagator, too, is a Green function of the Dirac operator:

$$(i\not{\partial} - m)S_F(x-y) = \delta^4(x-y)$$

[9] I think it has something to do with the relation between the Dirac equation and the Klein-Gordon equation, see the section "Klein-Gordon strikes back"

22 Interactions and perturbation Theory

Until now the Lagrangian of the fields has been free, i.e. without a potential term: it will now be introduced alongside the kinetic and mass terms.

The interaction term is

$$H_{\text{int}} := \int \mathrm{d}^3x \mathcal{H}_{\text{int}}(\phi(x^\mu)) = -\int \mathrm{d}^3x \mathcal{L}_{\text{int}}(\phi(x^\mu))$$

Note The interaction term is independent of the derivatives of the field: this is not required, but it is easier to work with, since adding this dependency would complicate the expression of the conjugate fields.

Note All theories must be *renormalizable*, i.e. the physical quantities which most of the time turn out to be non-converging integrals must be tamed: one of the ways is to swap ∞ in the extrema of integration with a "large" parameter Λ, solve that and hope the result is not dependent on Λ; this happens when Λ is a dimensionless parameter (e.g. a coefficient of ϕ^4 in a scalar interaction).

Other kinds of interaction are the *Yukawa* interaction, which introduces a term

$$\mathcal{H}_{\text{int}} = g\bar{\phi}\psi;$$

and the *electromagnetic interaction* of the QED theory

$$\mathcal{H}_{\text{int}} = e\bar{\psi}\gamma^\mu A_\mu \psi.$$

The problem of interacting theories is that **nothing can be solved exactly beyond the ϕ^4 interaction.**

22.1 Perturbation in Quantum Mechanics

The strategy to solve this complex problems is to consider a Hamiltonian with known solutions, e.g. the free Hamiltonian, and add to it a "small" perturbation: these are called *weakly interacting theories*.

Going back to QM, the propagator is

$$K(x_f, x_i, t_f, t_i) = \langle x_f | \, U(t_f, t_i) \, | x_i \rangle$$

and, as is known, it is a Green function G of the Schrödinger operator

$$(E - H)G = \mathbb{I}$$

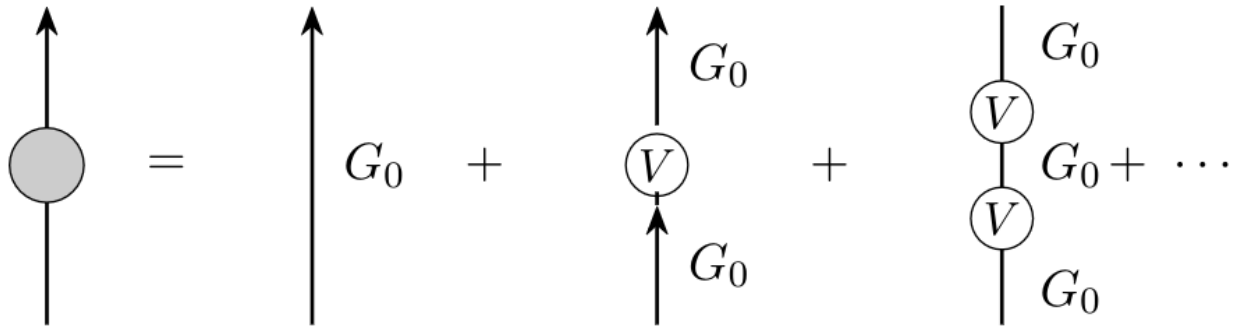
and so it can be, formally, written as

$$G = \lim_{\varepsilon \to 0} \frac{1}{E - H + i\varepsilon} = \frac{1}{E - H_0 - V} = \frac{1}{E - H_0 \left(1 - \frac{V}{E - H_0}\right)}$$

In this way, the problem can be expanded in power series via the geometric series

$$G = \frac{1}{E - H_0} \frac{1}{1 - \frac{V}{E - H_0}} = G_0 \left(1 + VG_0 + VG_0VG_0 + ...\right)$$

where G_0 is the propagator of the free Hamiltonian. This is called **Dyson series** and can be represented graphically.



22.2 Perturbation in QFT

The free propagator is defined as

$$G_0 = \langle 0| T\phi(x)\phi^\dagger(y) |0\rangle$$

while the most general, interacting propagator is

$$G = \langle \Omega| T\phi(x)\phi^\dagger(y) |\Omega\rangle$$

where $|\Omega\rangle$ is *the ground state of the interacting system* and T is the usual time-ordering operator: its interpretation is the same as the free propagator's with $|\Omega\rangle$ in place of $|0\rangle$.

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[...time evolution operator and interacting field, sec 8.3.1 notes] In the interacting picture **the states evolve in time according to the interaction Hamiltonian, while operators evolve according to the free Hamiltonian .** Time dependency has a *reference time* t_0, in which the regular QM Heisenberg's picture coincide with the interaction picture (i.e. before the perturbation starts).

The field in the interaction picture is given by

$$\phi_I(t, \mathbf{x}) := e^{iH_0(t-t_0)}\phi(t_0, \mathbf{x})e^{-iH_0(t-t_0)}$$

and all other operators are similarly defined, like the interaction Hamiltonian

$$H_I := e^{iH_0(t-t_0)}H_{\text{int}}e^{-iH_0(t-t_0)}$$

where H_{int} is built with the fields $\phi(t_0, \mathbf{x})$.

Inverting the relation between ϕ, ϕ_I yields

$$\phi(t, \mathbf{x}) := e^{-iH_0(t-t_0)}\phi_I(t_0, \mathbf{x})e^{iH_0(t-t_0)}.$$

Note The interacting field ϕ_I is a *free field*, because it is the time evolution of a free field with a free Hamiltonian: it can be interpreted as the evolution of the initial condition $\phi(t_0, \mathbf{x})$ with the interacting Hamiltonian switched off.

The full field $\phi(t, \mathbf{x})$ is defined as

$$\phi(t, \mathbf{x}) = e^{iH(t-t_0)}\phi(t_0, \mathbf{x})e^{-iH(t-t_0)}{}^{10}$$

[10] non sono sicuro, se H(t) posso usare la forma exp????

inserting the expression of $\phi(t_0)$ in terms of $\phi_I(t)$ yields

$$\phi(t,\mathbf{x}) = \underbrace{e^{iH(t-t_0)}e^{-iH_0(t-t_0)}}_{:=U_I^\dagger(t,t_0)}\phi_I(t,\mathbf{x})\underbrace{e^{iH_0(t-t_0)}e^{-iH(t-t_0)}}_{:=U_I(t,t_0)}$$

where U_I is the **time evolution operator in the interacting picture**.

The derivative of U_I is

$$i\frac{\partial}{\partial t}U_I = e^{iH_0(t-t_0)}(H-H_0)e^{-iH(t-t_0)}$$

recalling $H - H_0 = H_{\text{int}}$ and inserting $\mathbb{I} = e^{=iH_0(t-t_0)}e^{iH_0(t-t_0)}$ yields

$$i\frac{\partial}{\partial t}U_I(t,t_0) = \underbrace{e^{iH_0(t-t_0)}H_{\text{int}}e^{-iH_0(t-t_0)}}_{H_I(t)}\underbrace{e^{iH_0(t-t_0)}e^{-iH(t-t_0)}}_{U_I(t,t_0)}$$

So finally

$$i\frac{\partial}{\partial t}U_I(t,t_0) = H_I(t)U_I(t,t_0)$$

This differential equation is not trivial to solve as in QM, since H is a time dependent operator and causality may become an issue, since

$$[H_I(t_1), H_I(t_2)] = 0$$

does not hold $\forall \mathbf{t_1}, \mathbf{t_2}$, so the time ordering operator is here needed.

The QM form of the solution is still correct:

$$U_I(t,t_0) = T\exp\left[-i\int_{t_0}^{t}\mathrm{d}t' H_I(t')\right] = \mathbb{I} + (-i)\int_{t_0}^{t}\mathrm{d}t' H_I(t') + \frac{(-i)^2}{2!}\int_{t_0}^{t}\mathrm{d}t_1\int_{t_0}^{t}\mathrm{d}t_2 T\left[H_I(t_1)H_I(t_2)\right] + ...$$

but a problem arises with $\frac{\partial}{\partial t}U_I(t,t_0)$, since the double (and higher order) products' derivatives have problems with commutators:

$$\frac{\partial}{\partial t}\left(\int_{t_0}^{t}\mathrm{d}t_1 H_I(t_1)\int_{t_0}^{t}\mathrm{d}t_2 H_I(t_2)\right) = H_I(t)\int_{t_0}^{t}\mathrm{d}t'' H_I(t'') + \int_{t_0}^{t}\mathrm{d}t' H_I(t')$$

to reorder the second term the commutator between $H_I(t)$ and $H_I(t')$ must be null. To guarantee this, the *time ordering operator* is put at the start of the series expansion
!!!RIVEDI RIFAI ETC!!! PROVLEMA CON 2! ed estremi e T
where $t_0 < t_2 < t_1 < t$ is guaranteed by T.

Two useful properties of the time evolution operator are

$$U_I(t_1,t_2)U_I(t_2,t_3) = U_I(t_1,t_3),\ \forall t_1 \geq t_2 \geq t_3$$

$$U_I(t_1,t_3)U_I^\dagger(t_2,t_3) = U_I(t_1,t_2),\ \forall t_1 \geq t_2 \geq t_3$$

22.3 Ground and vacuum

The vacuum state is so that

$$H_0\left|0\right\rangle = 0$$

while the ground state of an interacting theory

$$H\left|\Omega\right\rangle = E_0\left|\Omega\right\rangle$$

with, in general, $E_0 \neq 0$.

Consider the complete set of eigenstates of the full Hamiltonian

$$|n\rangle \to \sum_n |n\rangle \langle n| = \mathbb{I}$$

and evolve the vacuum state up to a time $T > 0$

$$e^{-iHT}|0\rangle = e^{-iHT}\sum_n |n\rangle\langle n|0\rangle = \sum_n e^{-iE_nT}|n\rangle\langle n|0\rangle$$

since $|\Omega\rangle = |n=0\rangle$ and the interaction is small, there is an overlap of $|0\rangle$ and $|\Omega\rangle$:

$$e^{-iHT}|0\rangle = e^{-iE_0T}|\Omega\rangle\langle\Omega|0\rangle + \sum_{n\neq 0} e^{-iE_nT}|n\rangle\langle n|0\rangle\,.$$

In the limit $T\to\infty$, the first term is dominant, since $E_0 \leq E_n$. The formal way to do this is to "tilt" the limit of $T\to\infty$ to

$$T \to (\infty(1-i\varepsilon))$$

And so taking the limit, neglecting $n>0$ terms and inverting the last relation

$$|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \left(e^{-iE_0T}\langle\Omega|0\rangle\right)^{-1} e^{-iHT}|0\rangle$$

Since T goes to ∞, adding a constant to it does not change anything:

$$T \to T + t_0 = T - (-t_0)$$

So

$$|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \left(e^{-iE_0(T-(-t_0)}\langle\Omega|0\rangle\right)^{-1} e^{-iH(T-(-t_0))}|0\rangle$$

substituting now

$$|0\rangle = e^{iH_0(t_0-(-T))}|0\rangle$$

yields

$$|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \left(e^{-iE_0(T-(-t_0)}\langle\Omega|0\rangle\right)^{-1} e^{-iH(T-(-t_0))}e^{iH_0(t_0-(-T))}|0\rangle$$

so finally

$$\boxed{|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \left(e^{-iE_0(t_0-(-T)}\langle\Omega|0\rangle\right)^{-1} U_I(t_0,-T)|0\rangle}$$

which means that the ground state $|\Omega\rangle$ can be found by evolving the vacuum state.

$$\boxed{\langle\Omega| = \lim_{T\to\infty(1-i\varepsilon)} \langle 0|U_I(T,t_0)\left(e^{-iE_0(T-t_0)}\langle\Omega|0\rangle\right)^{-1}}$$

22.4 Gell-Mann-Low Formula

So now the full propagator formula can be derived from its definition [FORMATTAZIONE]

$$\langle\Omega|\phi(x)\phi(y)|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \left(e^{-iE_0(T-t_0)}\langle\Omega|0\rangle\right)^{-1} \langle 0|U_I(T,t_0)U_I^\dagger(x^0,t_0)\phi_I(x)U_I(x^0,t_0)U_I^\dagger(y^0,t_0)\phi_I(y)$$
$$U_I(y^0,t_0)U_I(t_0,-T)|0\rangle\left(e^{-iE_0(t_0-(-T)}\langle\Omega|0\rangle\right)^{-1}$$

by the properties of the time evolution operator, some products collapse and the constants can be simplified by noting

$$1 = \langle\Omega|\Omega\rangle = \left(|\langle 0|\Omega\rangle|^2 e^{-iE_0 2T}\right)^{-1} \langle 0| U_I(T,t_0)U_I(t_0,-T)|0\rangle$$

from which

$$\left(|\langle\Omega|0\rangle|^2 e^{-iE_0 2T}\right)^{-1} = \frac{1}{\langle 0| U_I(T,-T)|0\rangle}.$$

The final expression, also called the *reduction formula*, is

$$\boxed{\langle\Omega| T\phi(x)\phi(y)|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \frac{\langle 0| T\left\{\phi_I(x)\phi_I(y)\exp\left[-i\int_{-T}^{T} \mathrm{d}t H_I(t)\right]\right\}|0\rangle}{\langle 0| T\left\{\exp\left[-i\int_{-T}^{T} \mathrm{d}t H_I(t)\right]\right\}|0\rangle}}$$

This formula has an elegant expression in terms of the S-matrix

$$S := \lim_{T\to\infty(1-i\varepsilon)} T\left\{\exp\left[-i\int_{-T}^{T} \mathrm{d}t H_I(t)\right]\right\}$$

which is

$$\boxed{\langle\Omega| T\phi(x)\phi(y)|\Omega\rangle = \frac{\langle 0| T\{\phi_I(x)\phi_I(y)S\}|0\rangle}{\langle 0| S|0\rangle}}$$

22.5 Wick's Theorem

This theorem is very important, because it relates *time ordering* with *normal ordering* (which is easier to compute on the vacuum state).

The first step is to define two separate parts of a scalar field

$$\phi(x) = \phi^+(x) + \phi^-(x)$$

where

$$\phi^-(x) \propto \int \mathrm{d}^3k a_k,\ \phi^+(x) \propto \int \mathrm{d}^3k a_k^\dagger.$$

So the product

$$\phi(x)\phi(y) = \phi^+(x)\phi^+(y) + \underbrace{\phi^-(x)\phi^+(y)}_{\text{not ordered}} + \phi^+(x)\phi^-(y) + \phi^-(x)\phi^-(y)$$

is almost normal ordered

$$N(\phi(x)\phi(y)) = \phi^+(x)\phi^+(y) + \phi^+(y)\phi^-(x) + \phi^+(x)\phi^-(y) + \phi^-(x)\phi^-(y)$$

Supposing $x^0 > y^0$, the difference between the time ordered and normal ordered product is

$$T\{\phi(x)\phi(y)\} - N\{\phi(x)\phi(y)\} = [...] = \left[\phi^-(x), \phi^+(y)\right]$$

and vice versa if $y^0 > x^0$: the notation for this is the **contraction** of the two fields

$$T\{\phi(x)\phi(y)\} - N\{\phi(x)\phi(y)\} := \overline{\phi(x)\phi(y)} := \begin{cases} \left[\phi^-(x), \phi^+(y)\right] \text{ if } x^0 > y^0 \\ \left[\phi^-(y), \phi^+(x)\right] \text{ if } y^0 > x^0 \end{cases}$$

Lesson 22, 02/11

[11] **Note** The contraction of two fields is a *complex number*, for the commutation relations of $a, a^\dagger$ (show it!), so

$$\boxed{T\{\phi(x)\phi(y)\} = N\left\{\phi(x)\phi(y) + \overline{\phi(x)\phi(y)}\right\}}$$

With this result, the free propagator's expression greatly simplifies to

$$\langle 0|\, T\{\phi(x)\phi(y)\}\, |0\rangle = \underbrace{\langle 0|\, N\{\phi(x)\phi(y)\}\, |0\rangle}_{:=0} + \langle 0|\, \overline{\phi(x)\phi(y)}\, |0\rangle = \overline{\phi(x)\phi(y)} = i\Delta_F(x-y)$$

The generalization of this result is called **Wick's theorem**:
Wick's theorem: *The time ordered product of fields is equal to the sum of the normal ordering of the product with all possible contractions between the fields:*

$$T\left\{\prod_i \phi(x_i)\right\} = N\left\{\prod \phi_i(x_i)\right\} + \sum_{ij} \overline{\phi(x_i)\phi(x_j)}$$

Note The contractions, as seen before, are all the possible Feynman free propagators of the fields.

Lesson 23, 03/11

Proof of Wick's theorem (by induction)
The base case is with two fields and it is easily proven by brute force.

Supposing the theorem holds for n fields, consider the time-ordered product of $n+1$ fields: without losing generality, consider $x^0_{n+1} < x^0_i$, then

$$T\{\phi(x_1)...\phi(x_{n+1})\} = T\{\phi(x_1)...\phi(x_n)\}\,\phi(x_{n+1})$$

from the hypothesis, Wick's theorem can be used

$$T\{\phi(x_1)...\phi(x_{n+1})\} = \left(N\{\phi(x_1)...\phi(x_n)\} + \sum \text{all contractions}\right)\phi(x_{n+1}).$$

Consider now

$$N\{\phi(x_1)...\phi(x_n)\}\left(\phi(x_{n+1})^+ + \phi(x_{n+1})^-\right).$$

Noting that the normal-ordered product can be written as

$$N\{\phi(x_1)...\phi(x_n)\} = \sum_{A,B}\prod_{i\in A}\phi^+(x_i)\prod_{j\in B}\phi^-(x_j)$$

(where A, B are partitions of the set of indexes) yields

$$N\{\phi(x_1)...\phi(x_n)\}\,\phi(x_{n+1}) = \sum_{A,B}\prod_{i\in A}\phi^+(x_i)\prod_{j\in B}\phi^-(x_j)\left(\phi^+(x_{n+1}) + \phi^-(x_{n+1})\right).$$

The $\phi^-(x_{n+1})$ is not problematic, since it is already normal ordered, but in order to express this normal-ordered product in terms of $N\{\phi(x_1)...\phi(x_{n+1})\}$, the $\phi^+(x_{n+1})$ needs to go through all the $\phi^-(x_j)$'s yielding commutators:

$$N\{\phi(x_1)...\phi(x_n)\}\,\phi(x_{n+1}) = N\{\phi(x_1)...\phi(x_{n+1})\} + \sum_{A,B}\prod_{i\in A}\phi^+(x_i)\phi^+(x_{n+1})\prod_{j\in B}\phi^-(x_j) +$$
$$+\sum_{A,B}\prod_{k\in B}\phi^+(x_i)\prod_{j\neq k\in B}\phi^-(x_j)\overline{\phi(x_k)\phi(x_{n+1})}$$

[11] most of this lesson was a review of the previous online lesson

So the contractions sum with the others and the normal-ordered product now includes $\phi(x_{n+1})$ □

Example: four fields
Denoting $\phi_i = \phi(x_i)$, consider

$$T\{\phi_1\phi_2\phi_3\phi_4\} = N\left\{\overline{\phi_1\phi_2}\phi_3\phi_4 + \overline{\phi_2\phi_3}\phi_1\phi_4 + \overline{\phi_3\phi_4}\phi_1\phi_2 + \overline{\phi_1\phi_4}\phi_2\phi_3 + \overline{\phi_1\phi_2}\,\overline{\phi_3\phi_4} + \overline{\phi_1\phi_4}\,\overline{\phi_2\phi_3} + \overline{\phi_1\phi_3}\,\overline{\phi_2\phi_4}\right\}$$

When taking the vacuum expectation value the first 4 terms vanish because of N:

$$\langle 0|\, T\{\phi_1\phi_2\phi_3\phi_4\}\,|0\rangle = i\Delta_F(x_1-x_2)i\Delta_F(x_3-x_4) + i\Delta_F(x_1-x_4)i\Delta_F(x_2-x_3) + i\Delta_F(x_1-x_3)i\Delta_F(x_2-x_4)$$

Note The proof for Dirac fields is similar, save a $(-1)^p$ sign, given by the fact that p anticommutation relations need to be used.

22.6 The S-matrix

This is the greatest success of Quantum Field Theory: the predictive part of the Standard Model has passed with flying colours every test yet devised; the core of the predictions is the S-matrix.

As said before, the S-matrix is defined as

$$\lim_{T\to\infty} U_I(T,-T) = \lim_{T\to\infty} T\left\{\exp\left[-i\int_{-T}^{T} \mathrm{d}t H_I(t)\right]\right\} := S$$

Since the exponential is just notation for its series, the S-matrix can also be written as

$$S = T\left\{\mathbb{I} - i\int \mathrm{d}^4x \mathcal{H}_I(x^\mu) + \frac{(-i)^2}{2!}\int \mathrm{d}^4y \int \mathrm{d}^4w \mathcal{H}_I(y^\mu)\mathcal{H}_I(w^\mu)\right\}$$

where $\mathcal{H}_I$ is the interacting picture Hamiltonian density.
The reason behind the $T \to \pm\infty$ is to study **scattering processes** by considering the particles as non-interacting in the initial and final state: e.g. the initial state may be

$$a^\dagger(\mathbf{k})a^\dagger(\mathbf{k}')\,|0\rangle$$

which is a state with 2 scalar *free* particles (as $T \to -\infty$) of momenta $\mathbf{k}, \mathbf{k}'$, while the final state may be

$$\langle 0|\, a(\mathbf{k}')a(\mathbf{k})$$

which are again *free* particles (as $T \to \infty$): the probability amplitude of this happening is

$$A = \langle \text{final}|\, S\,|\text{initial}\rangle :$$

the convenient thing is that the S-matrix can be computed perturbatively using the exponential series.
Note The order of the series is the *number of vertices* in a **Feynman Diagram**.

22.7 The ϕ^4 self-interaction

The simplest interaction Hamiltonian density is

$$\mathcal{H}_I = \frac{\Lambda}{4!}\phi^4(x)$$

where Λ is a parameter.

Lesson 24, 04/11

This interaction will be plugged into the S-matrix expression:

$$S = T\left\{\mathbb{I} - i\int \mathrm{d}^4x\mathcal{H}_I(x^\mu) + \frac{(-i)^2}{2!}\int \mathrm{d}^4y\int \mathrm{d}^4w\mathcal{H}_I(y^\mu)\mathcal{H}_I(w^\mu) + ...\right\},$$

the integrals are over all space-time, but, *under the hypothesis that the interaction is short-ranged in space and in time*, the integrand is non-zero in a finite region of space-time; this also means that for the double or triple integrals the dominion over which the integrand is non-zero becomes progressively smaller: this **guarantees the convergence of the Dyson series**.

After finding the expression for S, the transition amplitude between the asymptotical states needs to be computed:

$$A = \langle q|\, S\, |p\rangle$$

in this case the initial state is a particle with momentum p and the final with q.

The first example is the scattering of a single particle, so the initial state is $a_p^\dagger\,|0\rangle$ and the final state is $\langle 0|\, a_q$ (with a normalization due to the choice of the 4-momenta over the 3-momenta):

$$A = (2\pi)^3 2(E_pE_q)^{\frac{1}{2}}\,\langle 0|\, a_q S a_p^\dagger\,|0\rangle\,.$$

The 0-th order of S is simply $\mathbb{I}$:

$$A = (2\pi)^3 2(E_pE_q)^{\frac{1}{2}}\,\langle 0|\, a_q\mathbb{I}a_p^\dagger\,|0\rangle = \langle q|p\rangle = (2\pi)^3 2(E_pE_q)^{\frac{1}{2}}\delta^3(\mathbf{p}-\mathbf{q}) = (2\pi)^3 2E_p$$

so nothing happens at perturbative order 0. The corresponding Feynman diagram is a single straight line.

The first order of S is not trivial:

$$A = -i(2\pi)^3 2(E_pE_q)^{\frac{1}{2}}\,\langle 0|\, a_q T\left\{\int \mathrm{d}^4x\frac{\Lambda}{4!}\phi^4(x^\mu)\right\}a_p^\dagger\,|0\rangle =$$
$$= -i\frac{\Lambda}{4!}(2\pi)^3 2(E_pE_q)^{\frac{1}{2}}\int \mathrm{d}^4x\,\langle 0|\, a_q\phi(x)\phi(x)\phi(x)\phi(x)a_q^\dagger\,|0\rangle$$

where the time ordering is automatically satisfied, since the field is evaluated 4 times at the same point in space-time.

There are 6 fields (4 ϕ's and 2 a's), so Wick's theorem prescribes to take all possible contractions between them: the first way to do so is to contract the two a's and two couples of ϕ (minding the fact that there are 3 ways to contract them), the second is to contract each a with a ϕ and two ϕ's between them (minding again the combinatorics, there are 12 ways to do so).

- The first kind of contractions yield 3 terms in the form

  $$\langle 0|\, aa^\dagger\,|0\rangle\,\overline{\phi(x)\phi(x)}\;\overline{\phi(x)\phi(x)} \propto \delta^3(\mathbf{q}-\mathbf{p})i\Delta_F^2(0),$$

  each consisting of a line and of a "double" loop: they represent, respectively, the case of a non-interacting particle and **the vacuum creating and reabsorbing a particle and an antiparticle** and they will simplify with the denominator in the Gell-Mann-Low formula. The diagrams in this first class are called *disconnected*.
  Note the double loop integral is delicate, since

  $$\overline{\phi(x)\phi(x)}\;\overline{\phi(x)\phi(x)} = \int \frac{\mathrm{d}^4k\mathrm{d}^4k'}{(2\pi)^8}\frac{i}{k^2-m^2+i\varepsilon}\frac{i}{k'^2-m^2+i\varepsilon}$$

  will be integrated over d^4x, which yields the infinite volume of spacetime.

- The second kind of contractions yield 12 terms in the form

$$\overline{\phi(x)a_p^\dagger}\,\overline{\phi(x)\phi(x)}\,\overline{a_q\phi(x)}$$

each consisting, respectively, of a line *coming from infinity to the interaction point* x, a loop and a line *exiting from the interaction point* x *to infinity.* These loops are the self-interaction, which has the effect of modifying the mass of the particle by moving the poles in the propagators (this is renormalization, it will be treated in the Advanced QFT course).

$$\overline{\phi(x)a_p^\dagger} = \mathcal{N}\int \mathrm{d}^3k\,\langle 0|\,(a_k e^{-ikx} + a_k^\dagger e^{ikx})a_p^\dagger\,|0\rangle = [...] = \frac{e^{-ipx}}{(2\pi)^{\frac{3}{2}}2E_p}$$

and similarly

$$\overline{a_q\phi(x)} = \frac{e^{iqx}}{(2\pi)^{\frac{3}{2}}2E_q}.$$

The loop integral is a bit more delicate for the same reasons as the double loop.

The amplitude up to first order is in the form $A = A^{(0)} + A^{(1)}$. The $A^{(1)}$ term is composed of the two parts discussed earlier, denoted as $A_a^{(1)}$ and $A_b^{(1)}$:

$$A_a^{(1)} = (2\pi)^3 2E_p\delta^3(\mathbf{p}-\mathbf{q})\left(-\frac{i\Lambda}{8}\right)\int\frac{\mathrm{d}^4k\mathrm{d}^4k'}{(2\pi)^8}\frac{i}{k^2-m^2+i\varepsilon}\frac{i}{k'^2-m^2+i\varepsilon}\int\mathrm{d}^4x$$

$$A_b^{(1)} = \left(-\frac{i\Lambda}{2}\right)\int\frac{\mathrm{d}^4k}{2\pi^4}\frac{i}{k^2-m^2+i\varepsilon}\underbrace{\int\mathrm{d}^4x e^{i(q-p)x}}_{=(2\pi)^4\delta^4(q-p)} = \left(-\frac{i\Lambda}{2}\right)\int\mathrm{d}^4k\frac{i}{k^2-m^2+i\varepsilon}\delta^4(q-p).$$

22.7.1 Feynman Rules for self interaction

All these explicit calculations are *very difficult* (since this is the **simplest** interaction possible!), but there are rules by Feynman to simplify the matter of calculating the amplitude. These are related to spacetime representation.

1. Each vertex of interaction corresponds to a factor $-i\Lambda$ (in general, $-i$[coupling constant]);
2. Each internal line between x, y is a propagator and corresponds to a factor $i\Delta_F(x-y)$;
3. Each external line (i.e. a line with one vertex) is associated to a plane wave, a factor $e^{\pm ipx}$ ($+$ for outgoing and $-$ for incoming; for antiparticles it is the opposite);
4. Integrate the position of the vertices over all space-time;
5. Divide by the symmetry factor (the combinatorics from before).

There are correspondent rules in momentum space, which are **the ones actually used**, since they are more convenient.

1. Each vertex of interaction corresponds to a factor $-i\Lambda$ (in general, $-i$[coupling constant]);
2. Each internal line is a propagator and corresponds to a factor $i\tilde{\Delta}_F(k) = \frac{i}{k^2-m^2+i\varepsilon}$, denoted only by k;
3. Each external line (i.e. a line with one vertex) is associated to a factor 1;
4. Integrate over all loop momenta;
5. Use a normalization (...?)